


Lecture 12
Parametrised Quantum Circuits
VQE, Autoencoders, Born Machines and Classifiers

# Deep Learning in Classical and Quantum Physics

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# 1

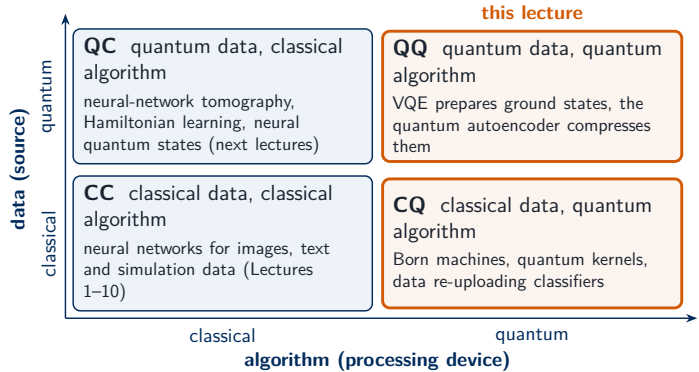
# Quantum Computers as Learning Models



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# Data and Algorithms Can Each Be Quantum

## Quantum computers as machine-learning models



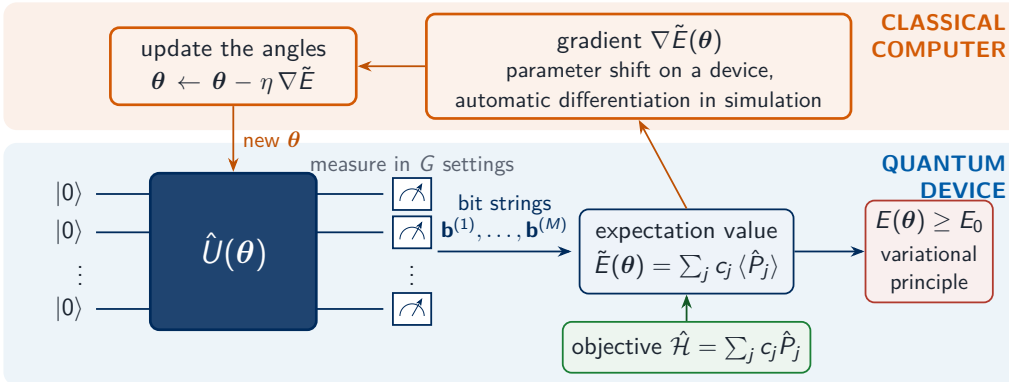

E. Aïmeur, G. Brassard and S. Gambs, LNCS **4013**, 431 (2006), doi:10.1007/11766247_37 V. Dunjko, J. M. Taylor and H. J. Briegel, PRL **117**, 130501 (2016), doi:10.1103/PhysRevLett.117.130501

# Five Routes Join Quantum Physics and Learning

- **Parametrised quantum circuits.** Trained gate angles give a trial wave function, a sampler or a classifier. **This lecture.**
- **Quantum kernels.** Overlaps of data-encoding states form a kernel for a support-vector machine. **This lecture.**
- **Quantum reservoir computing, extreme learning machines.** A fixed quantum system maps the input to measured features, and only a linear readout is trained (with memory for time series, or in a single pass).
- **Learning from quantum data.** A quantum memory holding copies of states learns some properties from exponentially fewer experiments.
- **Classical learning for quantum physics.** Neural networks reconstruct states, learn Hamiltonians and represent wave functions. **Next lectures.**

J. Biamonte *et al.*, Nature **549**, 195 (2017), doi:10.1038/nature23474 H.-Y. Huang *et al.*, Science **376**, 1182 (2022), doi:10.1126/science.abn7293 V. Havlíček *et al.*, Nature **567**, 209 (2019), doi:10.1038/s41586-019-0980-2 K. Fujii and K. Nakajima, Phys. Rev. Applied **8**, 024030 (2017), doi:10.1103/PhysRevApplied.8.024030 P. Mujal *et al.*, Adv. Quantum Technol. **4**, 2100027 (2021), doi:10.1002/qute.202100027 L. Innocenti *et al.*, Commun. Phys. **6**, 118 (2023), doi:10.1038/s42005-023-01233-w

# The Device Prepares, the Computer Optimises

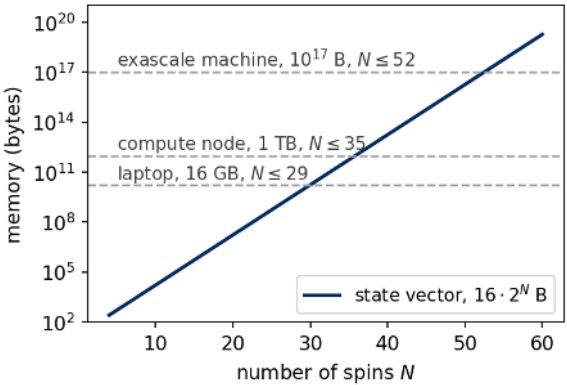



- The device prepares $|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0 \ldots 0\rangle$ and returns bit strings.
- A classical computer turns them into a cost and updates the angles. Every model of this lecture is this loop.

# Parametrised Circuits as Trainable Quantum Models

1. **Variational circuits** and their gradients.
2. **Ground states (VQE)** of qubit chains.
3. **Quantum autoencoders**: compress ground states, flag phase boundaries.
4. **QCNN and QGRNN**: classify phases, learn a Hamiltonian.
5. **Circuit autoencoder**: compress quantum channels.
6. **Born machines**: learn a probability distribution.
7. **Classifiers**: quantum kernels, data re-uploading.
8. **Limitations**: random states and barren plateaus.

# Memory Doubles with Every Added Qubit




implementation: see notebooks Memory $16 \cdot 2^N$ bytes of one state vector of $N$ qubits. Dashed lines mark a laptop, a compute node and an exascale machine.

- $N$ qubits: $2^N$ complex amplitudes. Each added qubit doubles the memory, so the largest computers store about fifty qubits.
- Classical approximations store far fewer numbers and miss some physics. A quantum register holds all $2^N$ amplitudes physically.

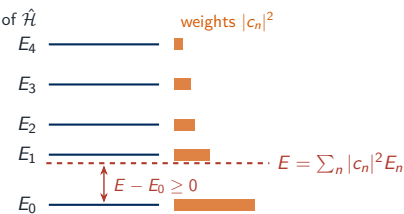
# Every Trial State Bounds the Ground Energy

## Variational principle

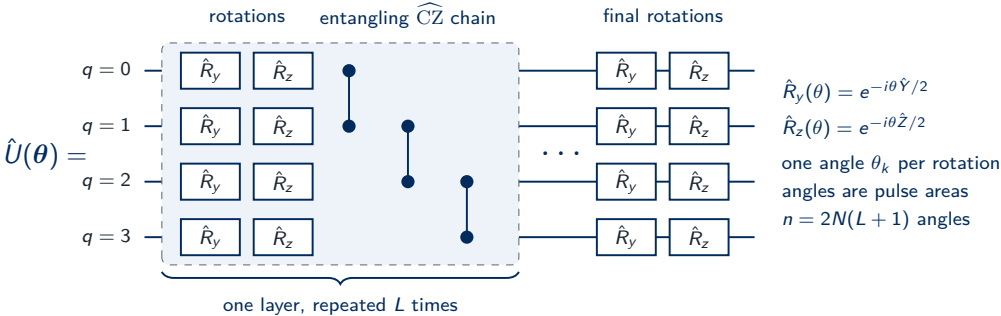
For every normalised state $|\psi\rangle$, $E[\psi] = \langle\psi|\hat{\mathcal{H}}|\psi\rangle \geq E_0$, with equality only for the ground state.

$$|\psi\rangle = \sum_m c_m |E_m\rangle \;\Rightarrow\; E = \sum_m \underbrace{|c_m|^2}_{\text{weights, sum } 1} \underbrace{E_m}_{\geq E_0} \geq E_0 \text{ levels of } \hat{\mathcal{H}}$$

- The energy is an average of the levels, never below the lowest.
- Minimise $E$ over a trial family $|\psi(\boldsymbol{\theta})\rangle$, the **ansatz**.



# Layers of Rotations and Entangling Gates Form the Circuit




- Each layer rotates every qubit by its own angles, then entangles neighbours with controlled-$Z$ gates.
- The angles are the trainable parameters, physically pulse areas.

# 2

# Parametrised Circuits and Their Gradients



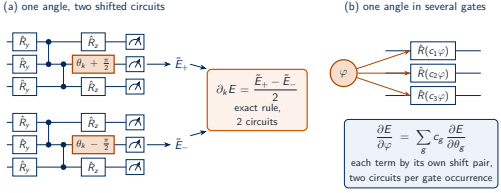
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# Two Shifted Circuits Give the Exact Gradient

$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle, \qquad \hat{U}(\boldsymbol{\theta}) = \hat{W}_n\, e^{-i\theta_n \hat{P}_n/2} \cdots \hat{W}_1\, e^{-i\theta_1 \hat{P}_1/2},$$

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_k)}{2}$$

- $\hat{P}_k$: Pauli string, $\hat{W}_k$: fixed gate. The energy $E(\boldsymbol{\theta}) = \sum_j c_j \langle \hat{P}_j \rangle$ is estimated from bit strings (VQE).
- $\hat{P}_k^2 = \hat{I}$ makes $E(\theta_k)$ a shifted cosine, so two circuits with $\theta_k \pm \pi/2$ give the exact derivative: $2n$ circuits per gradient.




In simulation: automatic differentiation (`jax.grad`) and Adam. Fewer circuits per gradient (SPSA): backup slides.
K. Mitarai *et al.*, PRA **98**, 032309 (2018), doi:10.1103/PhysRevA.98.032309 M. Schuld *et al.*, PRA **99**, 032331 (2019), doi:10.1103/PhysRevA.99.032331

# 3

# The Variational Quantum Eigensolver



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# VQE Targets the Ising Chain Ground State

A circuit's trial state should approach the ground state $|E_0\rangle$ of $\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j$, here the open transverse-field Ising chain,

$$\hat{\mathcal{H}} = \underbrace{-J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1}}_{\text{neighbours aligned along } z} \quad \underbrace{- h \sum_{i=0}^{N-1} \hat{X}_i}_{\text{each qubit along } +x} , \qquad J = 1.$$

- The two terms do not commute, so the ground state is a superposition of many bit strings.
- Ordered along $z$ for $h < J$, paramagnet for $h > J$, transition at $h = J$.
- On a device the energy $E(\boldsymbol{\theta})$ is the only accessible cost, estimated from bit strings.

# Entanglement Across a Cut Is Counted in Bits

Recalled from Lecture 11. Cut the chain into a left part $A$ and a right part $B$.

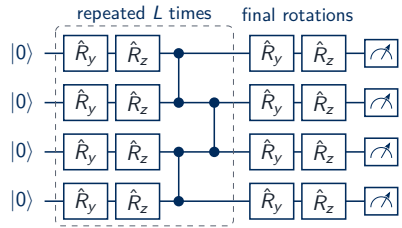
- Reshape the $2^N$ amplitudes into a $2^{N_A} \times 2^{N_B}$ matrix $\psi_{ab}$, rows labelled by the bits of $A$, columns by those of $B$.
- Its singular value decomposition gives the **Schmidt decomposition** with orthonormal $|a_k\rangle$, $|b_k\rangle$ and Schmidt coefficients $s_k > 0$,

$$|\psi\rangle = \sum_{k=1}^{\chi} s_k \, |a_k\rangle_A \otimes |b_k\rangle_B, \qquad S_{\mathrm{vN}} = -\sum_k s_k^2 \log_2 s_k^2 \quad \text{(in bits)}.$$

- A product state has one term, $S_{\mathrm{vN}} = 0$. A Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$ has $s_1^2 = s_2^2 = \tfrac{1}{2}$, $S_{\mathrm{vN}} = 1$ bit.
- The **half-chain entropy** uses the middle cut. Ansätze are compared by $S_{\mathrm{vN}}$, the energy error $\delta E = (E - E_0)/|E_0|$ and the fidelity $F = |\langle E_0|\psi\rangle|^2$, exact in simulation.

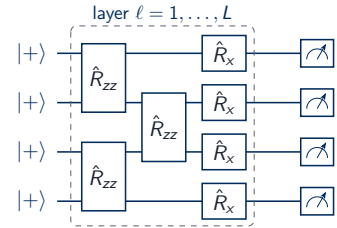
# Native Gates Versus Hamiltonian-Built Layers

## Hardware-efficient ansatz (HEA)




- $L$ layers of $\hat{R}_y\hat{R}_z$ rotations and a $\widehat{\mathrm{CZ}}$ chain, native gates.
- $n = 2N(L+1)$ independent angles, blind to $\hat{\mathcal{H}}$.

## Hamiltonian-variational ansatz (HVA)

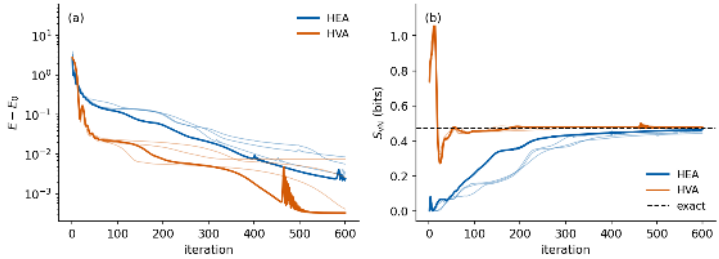



- Layers $e^{-i\beta_\ell \sum_i \hat{X}_i} e^{-i\gamma_\ell \sum_i \hat{Z}_i\hat{Z}_{i+1}}$ from the terms of $\hat{\mathcal{H}}$, like time steps of $e^{-i\hat{\mathcal{H}}t}$.
- $n = 2L$ shared angles, parity $\hat{\Pi} = \prod_i \hat{X}_i$ kept.

Our runs: $N = 6$ qubits, $L = 6$ layers, HEA 84 angles, HVA 12 angles.


A. Kandala *et al.*, Nature **549**, 242 (2017) D. Wecker *et al.*, PRA **92**, 042303 (2015) R. Wiersema *et al.*, PRX Quantum **1**, 020319 (2020)

# Energy and Mid-Chain Entanglement During Training



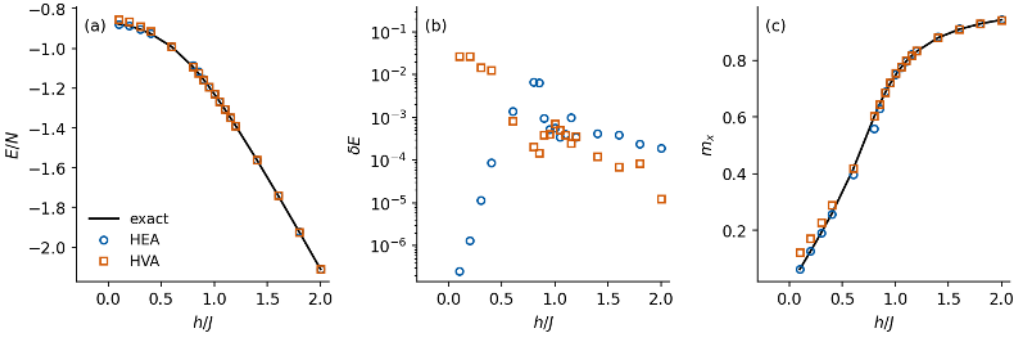

implementation: see notebooks

Critical Ising chain $h = J$, $N = 6$, $L = 6$, 4 restarts (thick: best). (a) $E - E_0$ in units of $J$, (b) middle-cut $S_{\mathrm{vN}}$, dashed exact, against Adam steps. On a device: Rényi-2 entropy $S_2 = -\log_2 \mathrm{Tr}\,\hat{\rho}_A^2 \leq S_{\mathrm{vN}}$ from randomized measurements.

- HVA (12 angles): smaller energy error, and the entropy reaches the exact value.
- HEA (84 angles): the energy falls first, the entanglement builds up slowly.

T. Brydges *et al.*, Science **364**, 260 (2019), doi:10.1126/science.aau4963

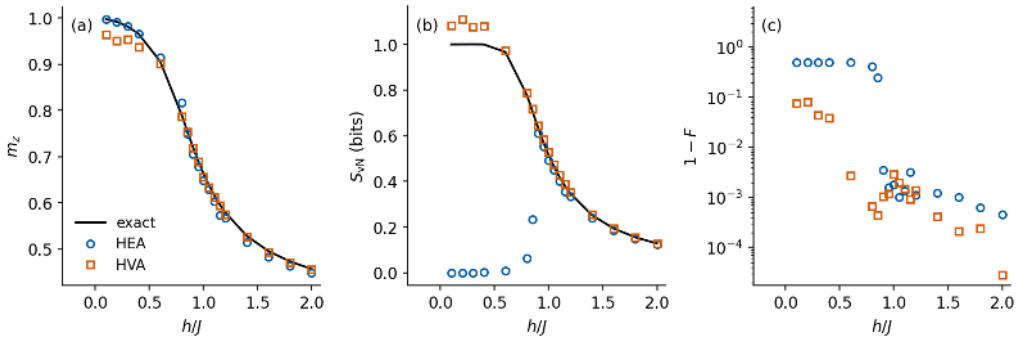
# VQE Energies Against Exact Results at $N = 8$




implementation: see notebooks

Exact state-vector simulations against $h/J$: (a) energy per site, (b) relative energy error $\delta E$, (c) transverse magnetisation $m_x$. Black lines exact diagonalisation, markers the best restart of each ansatz.

# VQE Order and Entanglement Against Exact Results at $N = 8$




implementation: see notebooks

Same runs: (a) order parameter $m_z$, (b) half-chain entropy $S_{\mathrm{vN}}$, (c) infidelity $1 - F$. Black lines exact diagonalisation, markers the best restart of each ansatz.

# Random States in Three Facts

A **uniformly (Haar-)random state** of $N$ qubits is a unit vector in $\mathbb{C}^d$, $d = 2^N$, drawn with no preferred direction.

1. **Tiny overlaps.** Its overlap with any fixed state is $|\langle\phi|\psi\rangle|^2 \approx 1/d = 2^{-N}$.
2. **Nearly maximal entanglement.** The entropy of $n_A \leq N/2$ qubits is close to $n_A$ bits, the **Page value**. Ground states of local Hamiltonians carry far less.
3. **Flat landscapes.** Deep random circuits produce such states. Gradients of a *global* cost, such as the fidelity with a target, then vanish exponentially in $N$, a **barren plateau**.

- **Local costs**, built from observables on one or a few qubits, keep larger gradients for shallow circuits.
- The last part derives these facts and measures their consequences.

D. N. Page, PRL **71**, 1291 (1993), doi:10.1103/PhysRevLett.71.1291 J. R. McClean *et al.*, Nat. Commun. **9**, 4812 (2018), doi:10.1038/s41467-018-07090-4

# 4

# Quantum Autoencoders



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# Quantum Data Need Compression Without Tomography

- A quantum simulation produces *quantum data*, e.g. VQE ground states, which exist only on the register.
- Storing or sending them costs qubits, and qubits are scarce.
- Classical compression would first need tomography, at a cost exponential in $N$.

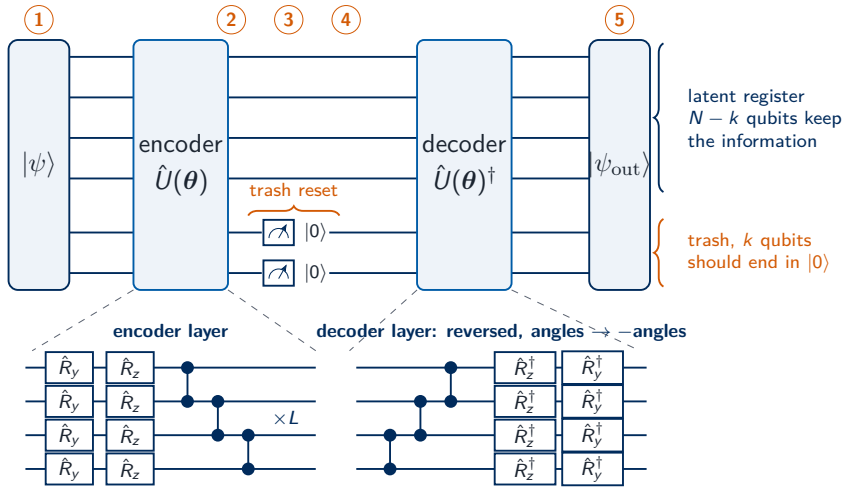
## Task

Given an ensemble $\{p_i, |\psi_i\rangle\}$ of states on $n+k$ qubits, find one circuit $\hat{U}(\boldsymbol{\theta})$ that moves all their information into $n$ latent qubits and leaves the other $k$ in a fixed reference state $|0\ldots0\rangle$, so that $\hat{U}^\dagger$ rebuilds every $|\psi_i\rangle$ from the latent qubits.

- A classical autoencoder squeezes $n+k$ bits through $n$, and a decoder restores them. Here the bottleneck is a smaller quantum register.

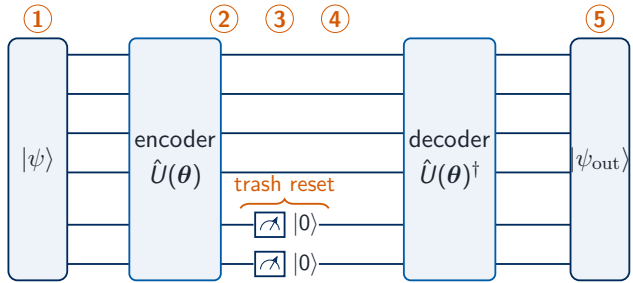
J. Romero *et al.*, Quantum Sci. Technol. **2**, 045001 (2017), doi:10.1088/2058-9565/aa8072

# An Encoder Squeezes States into Fewer Qubits




J. Romero *et al.*, Quantum Sci. Technol. **2**, 045001 (2017), doi:10.1088/2058-9565/aa8072

# The Autoencoder in Five Steps



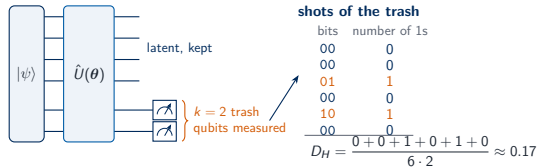

$$\underbrace{\hat{\rho}_{\text{in}} = |\psi_{\text{in}}\rangle\langle\psi_{\text{in}}|}_{\text{(1) input}} \xrightarrow{\hat{U}} \underbrace{\hat{\rho}_1 = \hat{U}\hat{\rho}_{\text{in}}\hat{U}^\dagger}_{\text{(2) encoded}} \xrightarrow{\text{Tr}_{\text{trash}}} \underbrace{\hat{\rho}_{\text{lat}} = \text{Tr}_{\text{trash}}\,\hat{\rho}_1}_{\text{(3) latent, } N-k \text{ qubits}}$$

$$\xrightarrow{\text{reset}} \underbrace{\hat{\rho}_2 = \hat{\rho}_{\text{lat}} \otimes |0\ldots0\rangle\langle 0\ldots0|_{\text{trash}}}_{\text{(4) fresh trash}} \xrightarrow{\hat{U}^\dagger} \underbrace{\hat{\rho}_{\text{out}} = \hat{U}^\dagger\hat{\rho}_2\,\hat{U}}_{\text{(5) output}} \overset{?}{\approx} \hat{\rho}_{\text{in}}$$

# A Clean Trash Register Means Lossless Compression

- Quality $F = \langle\psi_{\mathrm{in}}|\hat{\rho}_{\mathrm{out}}|\psi_{\mathrm{in}}\rangle$, ensemble average $\bar{F} = \sum_i p_i\, F(|\psi_i\rangle, \hat{\rho}_{\mathrm{out},i})$.
- If $\hat{U}|\psi_{\mathrm{in}}\rangle = |\psi^{\mathrm{c}}\rangle_{\mathrm{lat}} \otimes |0\ldots0\rangle_{\mathrm{trash}}$, then $\hat{\rho}_{\mathrm{lat}}$ is pure, the reset changes nothing and $F = 1$.
- Otherwise latent and trash are entangled, $\hat{\rho}_{\mathrm{lat}}$ is mixed and the reset loses information. Checking the trash after step (2) suffices to train the encoder.

# The Trash Hamming Distance Is a Local Cost




The Hamming distance counts the 1s in a trash bit string. Averaged over shots, per qubit,

$$D_H = \frac{1}{k} \sum_{j \in \text{trash}} \langle \hat{n}_j \rangle, \qquad \hat{n}_j = \frac{\hat{I} - \hat{Z}_j}{2}, \qquad F \geq \max\{0, 1 - kD_H\}^2.$$

- $D_H = 0$ means perfect reconstruction, $F = 1$. Only single-qubit counts enter: a local cost.
- $D_H = 0$ forces the trash into $|0 \ldots 0\rangle$, so it is *disentangled* from the latent qubits: $\operatorname{Tr} \hat{\rho}_B^2 \geq (1 - kD_H)^2$. The purity (Rényi-2 entropy $S_2 = -\log_2 \operatorname{Tr} \hat{\rho}_B^2$, SWAP test on a device) is the weaker test: any fixed pure trash state.

K. Kottmann *et al.*, PRR **3**, 043184 (2021), doi:10.1103/PhysRevResearch.3.043184 M. Cerezo *et al.*, Nat. Commun. **12**, 1791 (2021), doi:10.1038/s41467-021-21728-w

# Schmidt Rank Sets the Latent Qubit Count

The Schmidt decomposition across a cut $A|B$ (recalled in the VQE part) also gives the reduced state of $B$,

$$|\psi\rangle = \sum_{r=1}^{\chi} s_r \, |a_r\rangle_A \otimes |b_r\rangle_B, \qquad \hat{\rho}_B = \mathrm{Tr}_A |\psi\rangle\langle\psi| = \sum_{r=1}^{\chi} s_r^2 \, |b_r\rangle\langle b_r|.$$

- The number of terms $\chi$ is the **Schmidt rank**, equal to the rank of $\hat{\rho}_B$. $\chi = 1$ means a product state.

## Compress part $B$

An encoder acting on $B$ alone fits its information into $\lceil \log_2 \chi \rceil$ qubits and no fewer.

## One state

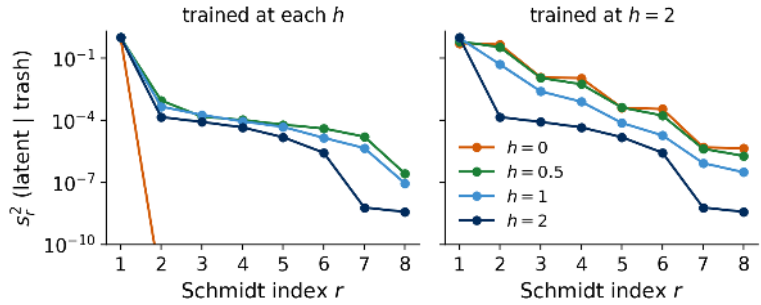
A global unitary always compresses it. A shallow one lowers $\chi$ by at most 2 per gate across a cut.

## A family of states

Compressible to $N - k$ qubits if the states span $\leq 2^{N-k}$ dimensions. Best fidelity: largest $2^{N-k}$ eigenvalues of the average state.

K. Fan, PNAS **35**, 652 (1949), doi:10.1073/pnas.35.11.652

# Exact Compression Means Schmidt Rank One

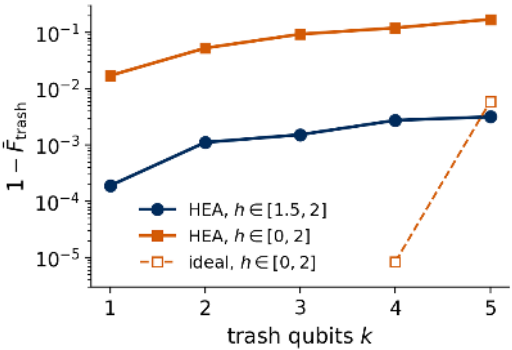



implementation: see notebooks

Squared Schmidt coefficients $s_r^2$ across the latent | trash cut, Ising chain, $N = 8$, $k = 3$.

- Left, encoder trained at each $h$: one term survives, $1 - s_1^2 \leq 10^{-3}$, so compression works.
- Right, encoder trained at $h = 2$ and frozen: the spectrum spreads in the ordered phase, $s_1^2 \approx 0.5$ at $h = 0$, and the reset loses information.

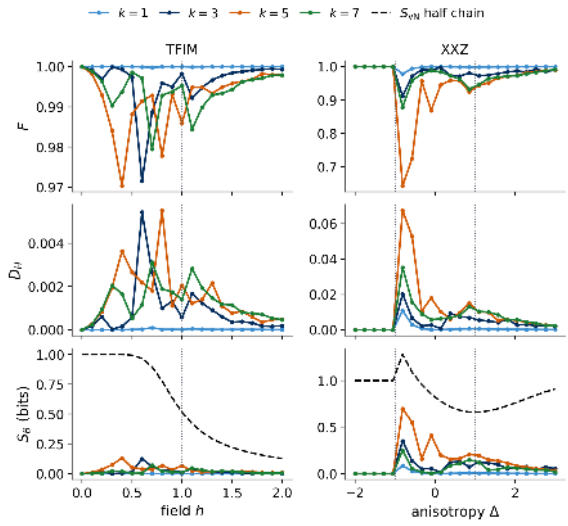
# Paramagnetic Ground States Compress to One Qubit




implementation: see notebooks Average trash infidelity after training against the number of trash qubits, for two families of ground states.

- Paramagnetic ground states ($h \in [1.5, 2]$) are close to one product state, so five of six qubits can be discarded with an error of a few $10^{-3}$.
- The family over $h \in [0, 2]$ crosses the transition. An ideal encoder still compresses it almost perfectly up to $k = 4$ (open squares).
- The three-layer HEA does not. The remaining error comes from the shallow circuit.

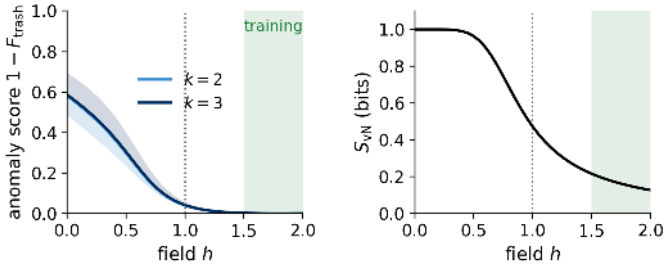
# Compression Is Hardest Where Entanglement Peaks




- A fresh autoencoder ($N = 8$, 4 layers, $D_H$ loss) per point, $k = 1, 3, 5, 7$ trash qubits.
- Rows: fidelity $F$, Hamming distance $D_H$, trash entropy $S_B$, exact half-chain entropy dashed.
- The Ising chain compresses well everywhere, $F \geq 0.97$, even the cat states of the ordered phase.
- The XXZ chain, $\hat{\mathcal{H}} = J\sum_i(\hat{X}_i\hat{X}_{i+1} + \hat{Y}_i\hat{Y}_{i+1} + \Delta\hat{Z}_i\hat{Z}_{i+1})$, is hardest just above $\Delta = -1$, where its entanglement peaks, and compresses perfectly in the ferromagnet $\Delta < -1$.

implementation: see notebooks

# Trash Infidelity Rises at the Phase Boundary



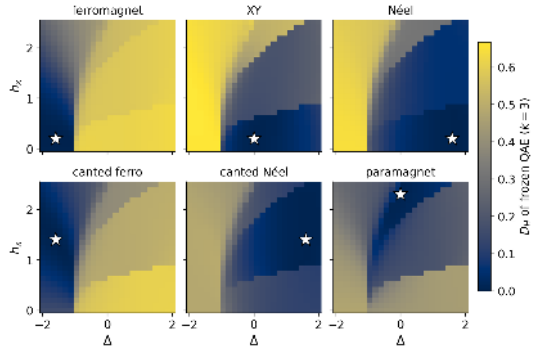

implementation: see notebooks

(a) Encoder trained on paramagnetic ground states (green band), frozen and applied at all fields, shading the spread over four starts. (b) Exact half-chain entropy.

- The trash infidelity is an **anomaly score**, near zero where the states resemble the training data. It rises with the entanglement when the chain orders, without labels.
- The position of the rise carries the physics, its height depends on the encoder.

K. Kottmann *et al.*, PRR **3**, 043184 (2021), doi:10.1103/PhysRevResearch.3.043184

# A Frozen Encoder Flags Other Phases

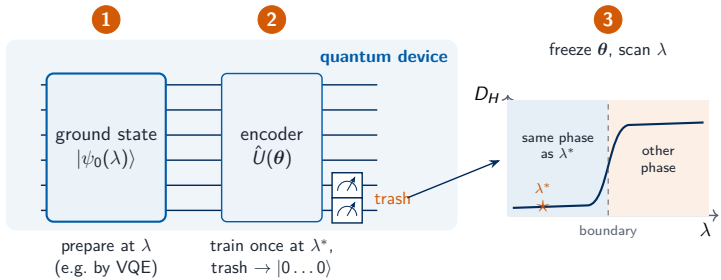



implementation: see notebooks

$\hat{\mathcal{H}} = J\sum_i(\hat{X}_i\hat{X}_{i+1} + \hat{Y}_i\hat{Y}_{i+1} + \Delta\hat{Z}_i\hat{Z}_{i+1}) + h_x\sum_i \hat{X}_i$, $N = 8$. Dark = compressed like the training state.

- Six autoencoders ($k = 3$), each trained on one ground state (star) and frozen.
- Every map isolates the region of its own reference, with borders along $\Delta = -1$ and the level crossings of the exact maps.

# Anomaly Detection Maps Phases Without Labels



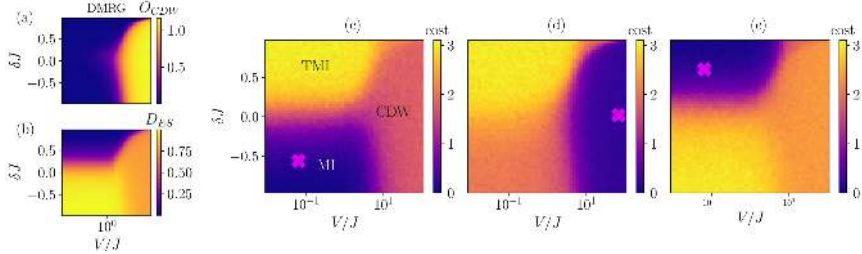

- The ground states never leave the device, with no tomography, no order parameter and no labels.
- Where the frozen encoder cannot clear the trash, a new phase begins.
- Retrain there and repeat until all boundaries are found.

K. Kottmann *et al.*, PRR **3**, 043184 (2021), doi:10.1103/PhysRevResearch.3.043184

# Anomaly Detection Finds a Topological Phase

Hard-core bosons, dimerised hopping, neighbour repulsion, $L = 12$ sites, one qubit per site,

$$\hat{H} = -\sum_i \left(J + \delta J(-1)^i\right)\left(\hat{b}_i^\dagger \hat{b}_{i+1} + \text{h.c.}\right) + V \sum_i \hat{n}_i \hat{n}_{i+1}.$$




(a) Charge-density-wave order parameter, (b) entanglement-spectrum degeneracy, both DMRG. (c)–(e) Anomaly syndrome trained on one state (cross) per phase, 6 trash qubits.

- Three phases appear, Mott insulator (MI), charge-density wave (CDW) and topological Mott insulator (TMI), whose edge states no local order parameter detects.

K. Kottmann *et al.*, PRR **3**, 043184 (2021), doi:10.1103/PhysRevResearch.3.043184

# 5
# QCNN and QGRNN



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# Quantum Convolutional Neural Networks for Phase Classification

- Anomaly detection found phase boundaries *without* labels. With labelled examples the same quantum data can be *classified*.
- A QCNN is a convolutional network written as a circuit: shared local gates as filters, measurements as pooling, one qubit as the output.
- Input is the ground state itself, on the device. Output is the phase, here a topological phase that no local measurement reveals.
- Depth and parameters grow only as $\log_2 N$.

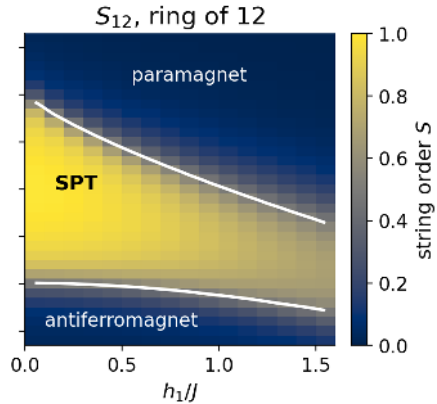
I. Cong, S. Choi and M. D. Lukin, Nat. Phys. **15**, 1273 (2019), doi:10.1038/s41567-019-0648-8

# Cluster-Ising Chain: Three Phases and a String Order

Model of Cong *et al.*, here on a ring of $N$ qubits, $J = 1$:

$$\hat{\mathcal{H}} = -J \sum_i \underbrace{\hat{Z}_{i-1}\hat{X}_i\hat{Z}_{i+1}}_{\hat{K}_i} - h_1 \sum_i \hat{X}_i - h_2 \sum_i \hat{X}_i\hat{X}_{i+1}$$

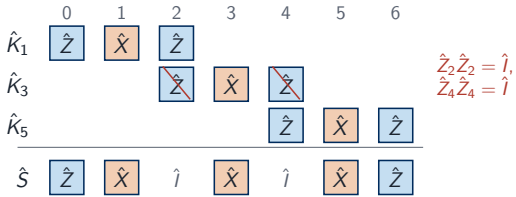
- Symmetry $\mathbb{Z}_2 \times \mathbb{Z}_2$: $\hat{X}_{\text{even}} = \prod_{i\,\text{even}} \hat{X}_i$, $\hat{X}_{\text{odd}} = \prod_{i\,\text{odd}} \hat{X}_i$.
- $h_1 = h_2 = 0$: all $\hat{K}_i$ commute, the ground state is the **cluster state** ($\hat{K}_i = +1$).
- $h_2 = 0$: exactly solvable (Jordan–Wigner), only non-local order distinguishes the phases.
- Phases: **SPT** (near the cluster state), **paramagnet** (large fields), **antiferromagnet** ($h_2 \lesssim -J$).




implementation: see notebooks Exact $\langle \hat{S} \rangle$, 12-site ring, white lines $\langle \hat{S} \rangle = 1/2$.

I. Cong *et al.*, Nat. Phys. **15**, 1273 (2019), doi:10.1038/s41567-019-0648-8

# A String of Operators Reveals the Topological Phase

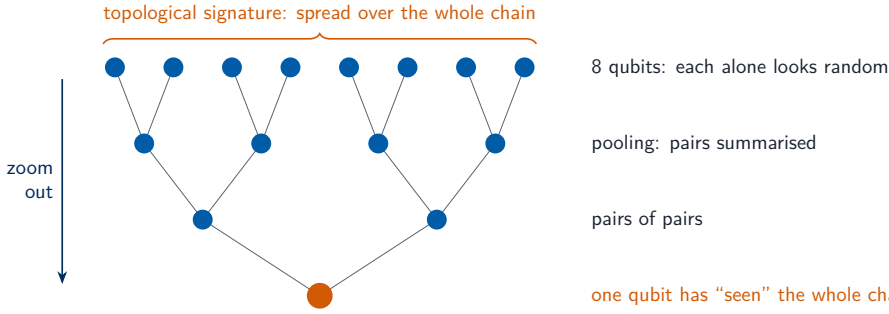



- The cluster state has $\hat{K}_i = +1$ for every $i$. Multiplying $\hat{K}_1\hat{K}_3\hat{K}_5$ cancels the inner $\hat{Z}$'s.
- The string $\hat{S} = \hat{Z}_0\hat{X}_1\hat{X}_3\hat{X}_5\hat{Z}_6$ therefore has $\langle\hat{S}\rangle = 1$, however long the string.
- No single-site or two-site observable separates the SPT phase from the paramagnet, only this long string does.

| | **SPT** | **paramagnet** | **antiferro.** |
|---|---|---|---|
| typical state | cluster state | $\lvert{+}{+}\cdots{+}\rangle$ | $\lvert{+}{-}{+}{-}\cdots\rangle$ |
| local order | none | $\langle\hat{X}_i\rangle \approx 1$, no symmetry broken | $\langle\hat{X}_i\hat{X}_{i+1}\rangle$ $-1$ |
| string $\langle\hat{S}\rangle$ | $\approx 1$ | $\approx 0$ | $\approx 0$ |
| open chain | edge modes, 4-fold degenerate | unique | 2 patterns |

F. Pollmann and A. M. Turner, PRB **86**, 125441 (2012), doi:10.1103/PhysRevB.86.125441

# Pooling Lets One Qubit See the Whole Chain




- The topological signature is a correlation along the chain. No single qubit, and no pair, shows it.
- The QCNN combines qubits step by step, like zooming out of an image: local details vanish, the overall pattern stays.
- Training chooses the gates so that the last qubit answers “topological” or “not topological”.

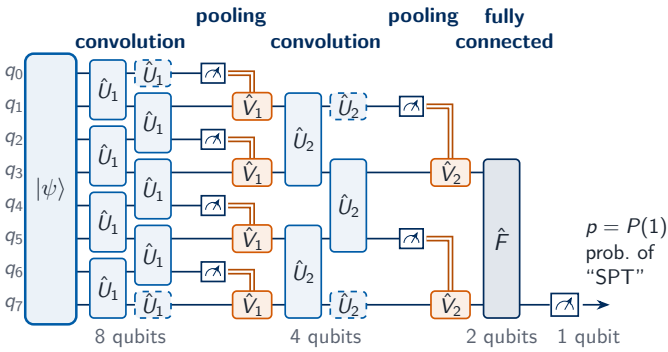
# Task: Classify the Phase of a Ground State

| Task | |
|---|---|
| **Input** | ground state $\|\psi(h_1, h_2)\rangle$ of the cluster-Ising chain, $N = 8$ qubits, prepared on the device |
| **Output** | one measured qubit, $p = P(1)$ is the probability of "SPT phase", class SPT if $p > 1/2$ |
| **Training data** | $M = 40$ states at known $(h_1, h_2)$, labels SPT where the exact $\langle \hat{S} \rangle > 1/2$ on a 12-site ring |
| **Trained** | the shared convolution, pooling and final unitaries, 57 angles |
| **Success** | accuracy of the predicted phase diagram on a grid, against the exact boundary |

- The supervised counterpart of anomaly detection, trained on labelled states instead of one reference state.
- The state never leaves the device. No tomography, one qubit is read out.

I. Cong, S. Choi and M. D. Lukin, Nat. Phys. **15**, 1273 (2019), doi:10.1038/s41567-019-0648-8

# QCNN: Shared Kernel, Pooling, Global Unitary




- **Convolution**: one two-qubit $\hat{U}_\ell$ on all neighbouring pairs, the shared *kernel* of a CNN.
- **Pooling**: measure one qubit per pair, apply $\hat{V}_\ell$ to its partner if the outcome is 1. The register halves.
- **Fully connected**: a global unitary $\hat{F}$, the dense layer. One qubit gives $p = P(1)$.
- Every gate is reused at all positions, 57 angles in total.

# Training Minimises the Mean Squared Error

- Output of the QCNN for an input state $|\psi\rangle$: the probability to read 1 on the output qubit,

$$p_{\boldsymbol{\theta}}(|\psi\rangle) = \langle\psi|\hat{W}(\boldsymbol{\theta})^{\dagger}\,|1\rangle\langle 1|_{\text{out}}\,\hat{W}(\boldsymbol{\theta})|\psi\rangle,$$

  with $\hat{W}(\boldsymbol{\theta})$ the whole circuit of convolutions, pooling and $\hat{F}$. On a device it is the frequency of 1 in repeated shots.

- Training set: $M = 40$ ground states $|\psi_\alpha\rangle$ with labels $y_\alpha = 1$ (SPT) or 0 (not SPT).

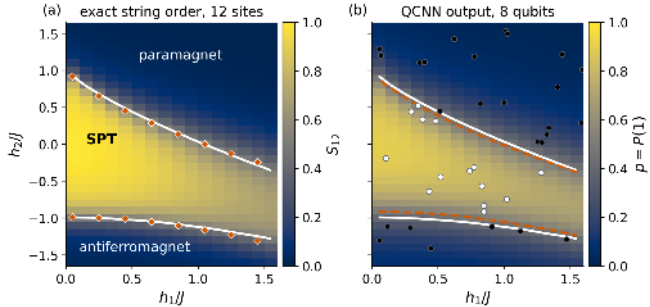
- Loss, the mean squared error between output and label,

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2M}\sum_{\alpha=1}^{M}\left[y_\alpha - p_{\boldsymbol{\theta}}(|\psi_\alpha\rangle)\right]^2.$$

- Gradients from the parameter-shift rule (device) or automatic differentiation (simulation), updates with Adam. A new state is classified SPT if $p_{\boldsymbol{\theta}} > 1/2$.

I. Cong *et al.*, Nat. Phys. **15**, 1273 (2019), doi:10.1038/s41567-019-0648-8

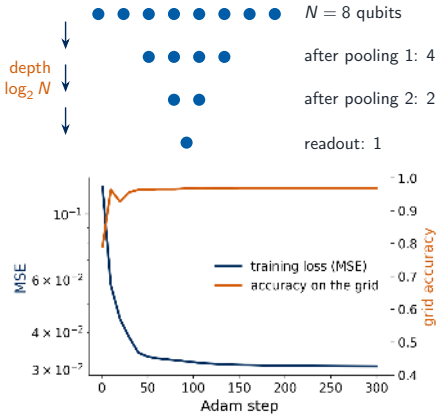
# The QCNN Finds the Topological Phase




implementation: see notebooks

- (a) Exact $S_{12}$, white line $S_{12} = 1/2$, diamonds where the ground state changes fastest. (b) QCNN output $p$, dashed $p = 1/2$, dots the training states.
- 97% of 528 grid points correct, as good as the rule $S_8 > 1/2$ on the same states (96%).
- At $h_1 = 0.5$ the learned boundary is $h_2 = 0.39$, the DMRG value of Cong *et al.* is 0.423.

# Logarithmic Depth Keeps the QCNN Trainable




- Only $\log_2 N$ short layers and one measured qubit: the gradients stay large, **no barren plateau**.
- Our run converges in about 50 Adam steps.

implementation: see notebooks

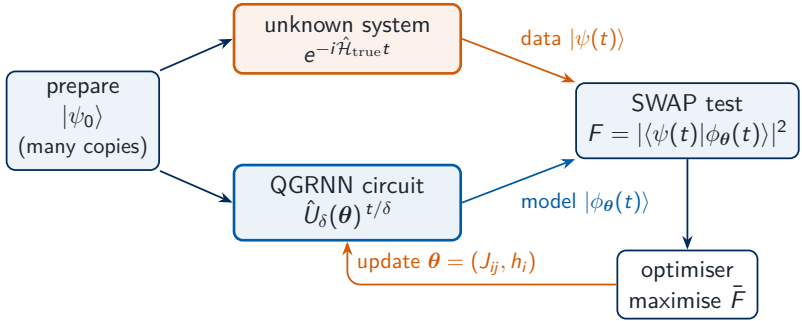
A. Pesah *et al.*, PRX **11**, 041011 (2021), doi:10.1103/PhysRevX.11.041011; I. Cong *et al.*, Nat. Phys. **15**, 1273 (2019), doi:10.1038/s41567-019-0648-8

# Learning a Hamiltonian from Quantum Data

- A quantum device or material evolves under a Hamiltonian $\hat{\mathcal{H}}$ we do not know: which qubits interact, and how strongly.
- We can prepare a state, let the system evolve for a time $t$, and keep the result. These states are **quantum data**.
- Idea: build a circuit that evolves states with a *trainable* Hamiltonian and tune it until it reproduces the data. The trained parameters are the learned $\hat{\mathcal{H}}$.
- This circuit is the **quantum graph recurrent neural network** (QGRNN).

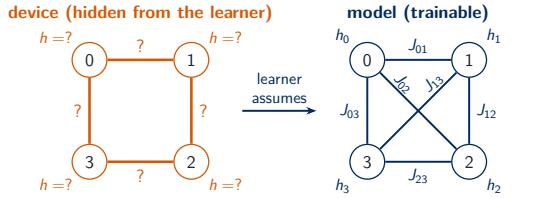
G. Verdon *et al.*, Quantum graph neural networks, arXiv:1909.12264 (2019)

# The QGRNN Loop at a Glance




- Copies of the same $|\psi_0\rangle$ go to the unknown system and to our trainable circuit.
- The SWAP test measures how close the two outputs are, the optimiser tunes $\boldsymbol{\theta}$ to bring them together.
- When they agree at many times $t$, the trained $J_{ij}, h_i$ are the hidden Hamiltonian.

# The Unknown Hamiltonian Lives on a Graph

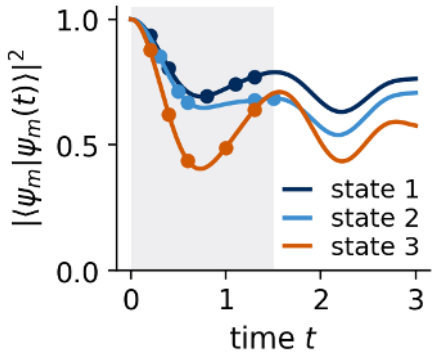



$$\hat{\mathcal{H}} = \underbrace{\sum_{(i,j)} J_{ij}\,\hat{Z}_i\hat{Z}_j}_{\hat{\mathcal{H}}_E,\ \text{edges}} + \underbrace{\sum_{i} \left(h_i\hat{Z}_i + g\,\hat{X}_i\right)}_{\hat{\mathcal{H}}_V,\ \text{nodes}}$$

- Node = qubit, edge = coupling $J_{ij} \neq 0$. Learning all six possible $J_{ij}$ therefore also learns the graph.
- Hidden: a ring of $N = 4$ qubits with its $J_{ij}$ and $h_i$. The transverse field $g = 1$ is known.

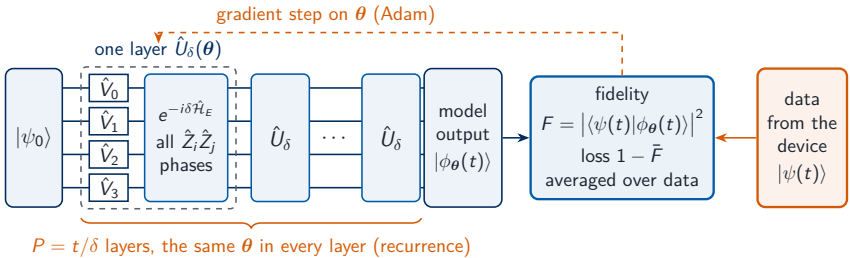
# Quantum Data: Initial States and Their Evolved Versions

- **Initial state** $|\psi_0\rangle$: a state we can prepare, here an approximate low-energy state of $\hat{\mathcal{H}}$.
- **Data state** $|\psi(t)\rangle = e^{-i\hat{\mathcal{H}}t}|\psi_0\rangle$: what the unknown system makes of $|\psi_0\rangle$ after time $t$.
- Repeating this for times $t_1, \ldots, t_K$ gives the data set of pairs $\big(|\psi_0\rangle, |\psi(t_k)\rangle\big)$. Here 3 initial states $\times$ 5 times.
- An exact eigenstate would only gain a phase and teach nothing, hence approximate ground states.




Overlap of each initial state with its evolved state, dots at the data times.

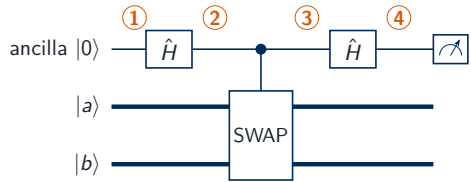
# The Model Repeats One Trainable Trotter Layer




- One layer $\hat{U}_\delta(\boldsymbol{\theta}) = e^{-i\delta\hat{\mathcal{H}}_E(\boldsymbol{\theta})}e^{-i\delta\hat{\mathcal{H}}_V(\boldsymbol{\theta})}$ evolves for a short time $\delta$, with trainable $\boldsymbol{\theta} = (J_{ij}, h_i)$.
- **Model state** $|\phi_{\boldsymbol{\theta}}(t)\rangle = \hat{U}_\delta(\boldsymbol{\theta})^{t/\delta}|\psi_0\rangle$: the same layer repeated $t/\delta$ times, which approaches $e^{-i\hat{\mathcal{H}}(\boldsymbol{\theta})t}$ for small $\delta$ (Trotter).
- The name: *graph* = which qubits interact, *recurrent* = one identical layer repeated, *neural network* = trainable $\boldsymbol{\theta}$.

H. F. Trotter, Proc. Am. Math. Soc. **10**, 545 (1959), doi:10.1090/S0002-9939-1959-0108732-6

# The SWAP Test Measures the Fidelity of Two States



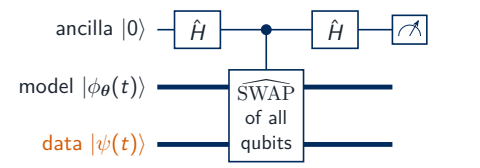

(1) $|0\rangle|a\rangle|b\rangle \xrightarrow{\hat{H}}$ (2) $\frac{1}{\sqrt{2}}\big(|0\rangle+|1\rangle\big)|a\rangle|b\rangle \xrightarrow{\text{controlled SWAP}}$ (3) $\frac{1}{\sqrt{2}}\big(|0\rangle|a\rangle|b\rangle+|1\rangle|b\rangle|a\rangle\big)$

$\xrightarrow{\hat{H}}$ (4) $\frac{1}{2}|0\rangle\big(|a\rangle|b\rangle+|b\rangle|a\rangle\big)+\frac{1}{2}|1\rangle\big(|a\rangle|b\rangle-|b\rangle|a\rangle\big)$

- Probability to read 0: $P(0)=\frac{1}{4}\big\||a\rangle|b\rangle+|b\rangle|a\rangle\big\|^2=\frac{1}{2}\big(1+|\langle a|b\rangle|^2\big)$.
- So $F=|\langle a|b\rangle|^2=2P(0)-1$, from repeated runs with fresh copies, without knowing $|a\rangle$, $|b\rangle$. Identical states always give 0.

H. Buhrman *et al.*, PRL **87**, 167902 (2001), doi:10.1103/PhysRevLett.87.167902

# Training Compares Data and Model States




$$P(\text{ancilla} = 0) = \frac{1+F}{2}, \qquad F = |\langle\psi(t)|\phi_{\boldsymbol{\theta}}(t)\rangle|^2$$

- Training step: pick a random data time $t_k$. One register holds the data state $|\psi(t_k)\rangle$, the other the model state $|\phi_{\boldsymbol{\theta}}(t_k)\rangle$.
- **SWAP test**: an ancilla in $|0\rangle$, Hadamard, controlled swap of the two registers, Hadamard, measure the ancilla.
- $P(\text{ancilla} = 0) = \frac{1}{2}(1+F)$ with the fidelity $F = |\langle\psi(t_k)|\phi_{\boldsymbol{\theta}}(t_k)\rangle|^2$.

$$\mathcal{L}(\boldsymbol{\theta}) = 1 - \overline{F}, \qquad \overline{F} = \text{average fidelity over the sampled data pairs}$$

H. Buhrman *et al.*, PRL **87**, 167902 (2001), doi:10.1103/PhysRevLett.87.167902

# Matching the Data Identifies the Hamiltonian

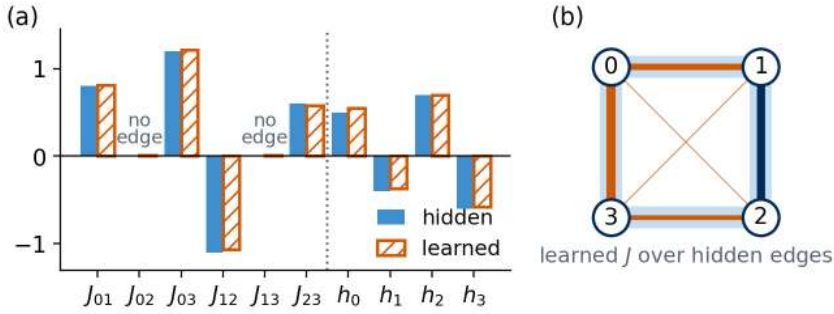
- Copies of $|\psi_0\rangle$ go both ways: one to the unknown system ($\to |\psi(t)\rangle$), one into our circuit ($\to |\phi_{\boldsymbol{\theta}}(t)\rangle$).
- The circuit starts with a coupling on *every* pair (complete graph) and random values. The optimiser raises $\overline{F}$ step by step.
- Why matching states fixes $\hat{\mathcal{H}}$:

$$e^{-i\hat{\mathcal{H}}t}|\psi_0\rangle = |\psi_0\rangle - it\,\hat{\mathcal{H}}|\psi_0\rangle - \tfrac{t^2}{2}\,\hat{\mathcal{H}}^2|\psi_0\rangle + \dots$$

  Agreement at many times fixes $\hat{\mathcal{H}}|\psi_0\rangle, \hat{\mathcal{H}}^2|\psi_0\rangle, \dots$, and several initial states fix $\hat{\mathcal{H}}$.

- Too little data is not enough: one data pair is fitted almost perfectly, yet the couplings are off by 0.18. With 15 pairs the error drops to 0.05.

# Training Recovers the Hidden Couplings and Graph

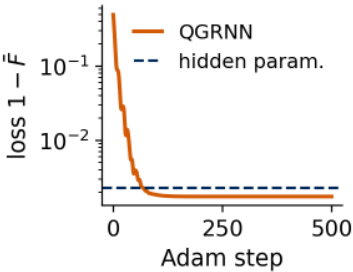



implementation: see notebooks

- From random couplings on all six pairs, Adam recovers every $J_{ij}$ and $h_i$ to within 0.05.
- The absent pairs $(0, 2)$, $(1, 3)$ end at $|J| \approx 0.01$: the learned graph is the hidden ring.

D. P. Kingma and J. Ba, arXiv:1412.6980

# The Trotter Step Limits the Accuracy







implementation: see notebooks

- The parameter error shrinks $\propto \delta$, at the price of $t/\delta$ layers.
- This task is **Hamiltonian learning**, continued in the next lecture.

G. Verdon *et al.*, arXiv:1909.12264

# 6

# Circuit Autoencoder



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# Circuits as Quantum Data

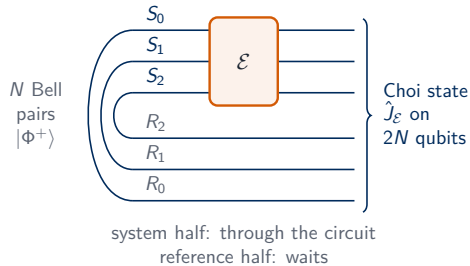
- The autoencoder compressed *states*. A circuit, or a black-box device, also stores information.
- A **quantum channel** maps density matrices to density matrices, $\mathcal{E}(\hat{\rho}) = \sum_a \hat{K}_a \hat{\rho} \hat{K}_a^\dagger$ with Kraus operators $\hat{K}_a$, $\sum_a \hat{K}_a^\dagger \hat{K}_a = \hat{I}$ (trace preserved).
- A unitary circuit has one Kraus operator. Dephasing, $\mathcal{E}(\hat{\rho}) = (1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}$, has $\hat{K}_0 = \sqrt{1-p}\,\hat{I}$, $\hat{K}_1 = \sqrt{p}\,\hat{Z}$.
- The Choi–Jamiołkowski isomorphism turns a channel into a state, so fidelity, spectrum and rank apply to channels.

## Task

Find encoders $\hat{U}, \hat{V}$ and a channel $\mathcal{F}$ on $m < N$ qubits that reproduce an unknown $N$-qubit channel $\mathcal{E}$, run as a black box.

J. Wu *et al.*, Phys. Rev. A **109**, 032623 (2024), doi:10.1103/PhysRevA.109.032623

# Choi–Jamiołkowski Isomorphism




- $N$ Bell pairs form $|\Phi^+\rangle = D^{-1/2}\sum_i |i\rangle_S |i\rangle_R$, system $S$, reference $R$, $D = 2^N$.
- The Choi state $\hat{J}_\mathcal{E} = (\mathcal{E}\otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$ is a density matrix on $2N$ qubits.
- It determines the whole channel, $\mathcal{E}(\hat{\rho}) = D\,\mathrm{Tr}_R\big[(\hat{I}\otimes \hat{\rho}^T)\hat{J}_\mathcal{E}\big]$ for every input $\hat{\rho}$.

M.-D. Choi, Linear Algebra Appl. **10**, 285 (1975), doi:10.1016/0024-3795(75)90075-0; A. Jamiołkowski, Rep. Math. Phys. **3**, 275 (1972), doi:10.1016/0034-4877(72)90011-0

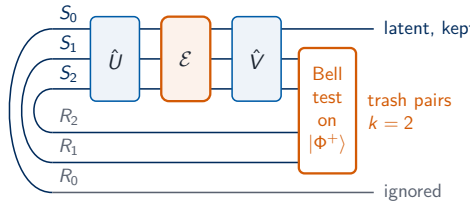
# Choi States of One-Qubit Channels

| channel $\mathcal{E}(\hat{\rho})$ | Choi state $\hat{J}_{\mathcal{E}}$ | eigenvalues | rank |
|---|---|---|---|
| identity, $\hat{\rho}$ | $\lvert\Phi^+\rangle\langle\Phi^+\rvert$ | $1, 0, 0, 0$ | 1 |
| Hadamard, $\hat{H}\hat{\rho}\hat{H}$ | pure | $1, 0, 0, 0$ | 1 |
| dephasing, $(1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}$ | mixed | $1-p,\ p,\ 0,\ 0$ | 2 |
| depolarising, $\hat{I}/2$ | $\hat{I}/4$ | $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$ | 4 |

- The rank of $\hat{J}_{\mathcal{E}}$ is the smallest number of Kraus operators $\hat{K}_a$ in $\mathcal{E}(\hat{\rho}) = \sum_a \hat{K}_a \hat{\rho} \hat{K}_a^\dagger$. Every unitary circuit has rank 1.
- Two channels are compared by the Choi fidelity $F(\hat{J}_{\mathcal{E}_1}, \hat{J}_{\mathcal{E}_2})$, an average over inputs.

implementation: see notebooks

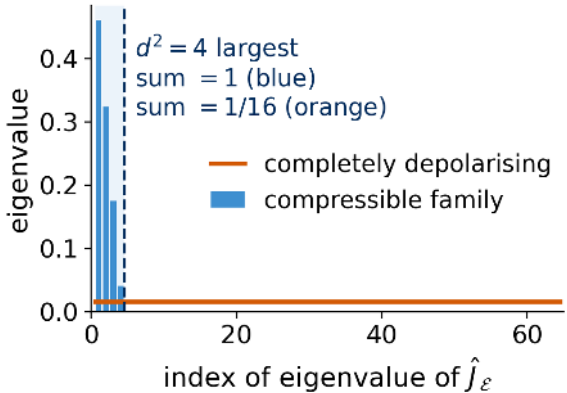
# Compressing a Circuit




- Encoders $\hat{U}$ before and $\hat{V}$ after $\mathcal{E}$. The channel $\mathcal{F}$ keeps what acts on the $m = N - k$ latent qubits.
- Rebuilt channel $\tilde{\mathcal{E}} = \hat{V}^\dagger \circ (\mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}) \circ \hat{U}^\dagger$, exact when $\hat{V} \circ \mathcal{E} \circ \hat{U} = \mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}$.
- The loss uses the trash alone, 1 − mean probability that a trash qubit and its reference pass a Bell test.

J. Wu *et al.*, Phys. Rev. A **109**, 032623 (2024), doi:10.1103/PhysRevA.109.032623

# Choi Spectrum Limits Compression

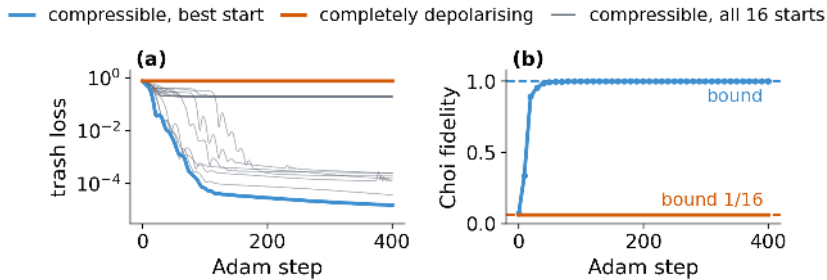



- $F(\hat{J}_{\mathcal{E}}, \hat{J}_{\tilde{\mathcal{E}}}) \leq$ sum of the $d^2$ largest eigenvalues of $\hat{J}_{\mathcal{E}}$, $d = 2^m$.
- Blue shows 6 circuits on $N = 3$ qubits acting on one hidden qubit, with bound 1 for $m = 1$.
- Orange shows complete depolarisation, $\hat{J}_{\mathcal{E}} = \hat{I}/64$, with bound $1/16$.

implementation: see notebooks

J. Wu *et al.*, Phys. Rev. A **109**, 032623 (2024), doi:10.1103/PhysRevA.109.032623

# Training on the Trash Qubits




- Hardware-efficient $\hat{U}(\theta)$, $\hat{V}(\varphi)$, 18 angles each, 16 random starts.
- Seven starts reach Choi fidelity above 0.999, the best $1 - 2 \times 10^{-5}$. Nine stop in local minima.
- The depolarising channel keeps loss 3/4 for every encoder, with zero gradient.

implementation: see notebooks

# 7

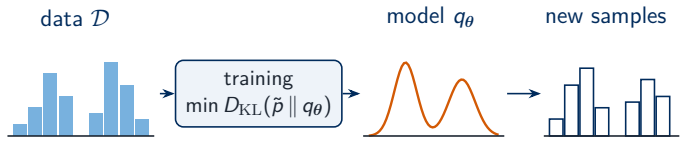
# Circuits as Probability Models



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# A Generative Model Samples Like the Data

- A data set $\mathcal{D}$ contains independent samples of an unknown distribution $p$.
- The goal is a model $q_\theta$ close to $p$ that can be sampled, whose new samples look like the data without copying them.




- A register measured in the computational basis returns Born-rule samples, so a parametrised circuit defines such a model.

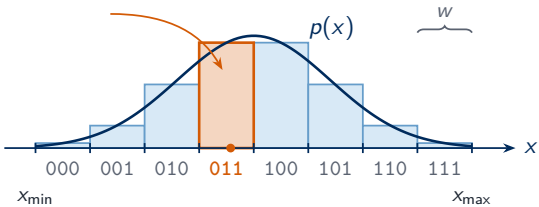
# A Wave Function Is a Probability Distribution

A state of $N$ qubits is a superposition of all $2^N$ bit strings,

$$|\psi\rangle = \sum_{\mathbf{b}\in\{0,1\}^N} \psi_{\mathbf{b}}\,|\mathbf{b}\rangle, \qquad q(\mathbf{b}) = \underbrace{|\psi_{\mathbf{b}}|^2}_{\text{Born rule}} \geq 0, \qquad \sum_{\mathbf{b}} q(\mathbf{b}) = \langle\psi|\psi\rangle = 1.$$

- The squared amplitudes form a normalised distribution over $2^N$ outcomes.
- Measuring every qubit draws one bit string $\mathbf{b}$ from $q$, so sampling is native to the device.
- A circuit $\hat{V}(\boldsymbol{\theta})$ with $n$ angles defines a family $q_{\boldsymbol{\theta}}$, and training moves probability between bit strings.
- In VQE the bit strings only estimate an energy, here their distribution is the model.

# Bit Strings Label Bins of a Continuous Variable

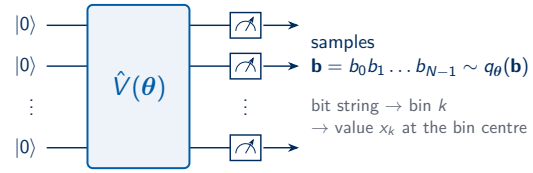



- $[x_{\min}, x_{\max}]$ is cut into $2^N$ bins of width $w$. Bin $k$ is the binary bit string of $k$, qubit 0 most significant,

$$k = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q}.$$

- A measured bit string is read back as the bin centre $x_k = x_{\min} + (k + \frac{1}{2})w$. Each added qubit halves $w$.
- Empirical distribution of the data, $\tilde{p}(\mathbf{b}) = n(\mathbf{b})/|\mathcal{D}|$, with $n(\mathbf{b})$ the count of $\mathbf{b}$ in $\mathcal{D}$.

# A Measured Circuit Is a Born Machine




A circuit $\hat{V}(\boldsymbol{\theta})$ on $|0\rangle^{\otimes N}$, then a measurement of every qubit, returns $\mathbf{b}$ with probability

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \left|\langle \mathbf{b}| \hat{V}(\boldsymbol{\theta}) |0\rangle^{\otimes N}\right|^2.$$

- A model whose probabilities are squared amplitudes is a **Born machine**.
- In simulation $q_{\boldsymbol{\theta}}$ is known exactly, a device gives only frequencies $n(\mathbf{b})/M$ over $M$ shots.

Z.-Y. Han *et al.*, PRX **8**, 031012 (2018), doi:10.1103/PhysRevX.8.031012 J.-G. Liu and L. Wang, PRA **98**, 062324 (2018), doi:10.1103/PhysRevA.98.062324 M. Benedetti *et al.*, npj Quantum Inf. **5**, 45 (2019), doi:10.1038/s41534-019-0157-8

# Minimal KL Divergence Means Maximal Likelihood

The Kullback–Leibler divergence (in nats) is non-negative and vanishes only for $q_\theta = p$,

$$D_{\mathrm{KL}}(p \,\|\, q_\theta) = \sum_{\mathbf{b}} p(\mathbf{b}) \ln \frac{p(\mathbf{b})}{q_\theta(\mathbf{b})}.$$

With the empirical $\tilde{p}$ in place of $p$, the sum weighted by $n(\mathbf{b})/|\mathcal{D}|$ becomes a sum over the data,

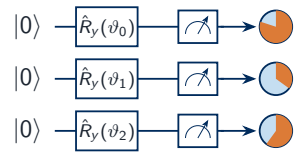
$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_\theta) = \underbrace{\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln \tilde{p}(\mathbf{b})}_{-H(\tilde{p}),\ \text{independent of } \theta} \underbrace{-\frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \ln q_\theta\big(\mathbf{b}^{(i)}\big)}_{\mathcal{L}_{\mathrm{NLL}}(\theta),\ \text{negative log-likelihood}} .$$

- Minimising the KL divergence and maximising the likelihood of the data are the same task with the same gradient.

S. Kullback and R. A. Leibler, Ann. Math. Statist. **22**, 79 (1951), doi:10.1214/aoms/1177729694

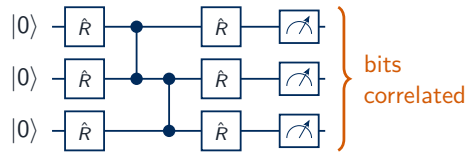
# Entangling Gates Correlate the Bits

(a) product circuit




independent biased coins
$q(\mathbf{b}) = \prod_q q_q(b_q)$, no correlations

(b) circuit with entanglers

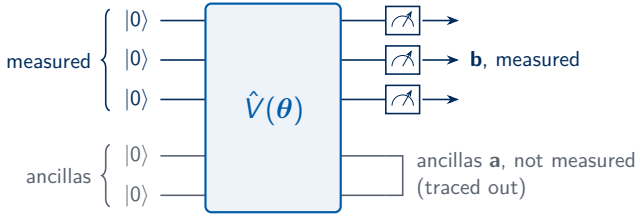



$\widehat{\mathrm{CZ}}$ gates create the correlations
that a two-peak target needs

best product model = product of the data marginals $p_q(b_q)$
$D_{\mathrm{KL}}$ never falls below the product bound

- Without entangling gates $q_\theta(\mathbf{b}) = \prod_q q_q(b_q)$, every bit an independent biased coin, so correlated data cannot be fitted.
- Entangling gates correlate the bits.

# Unmeasured Ancillas Turn the Model into a Mixture

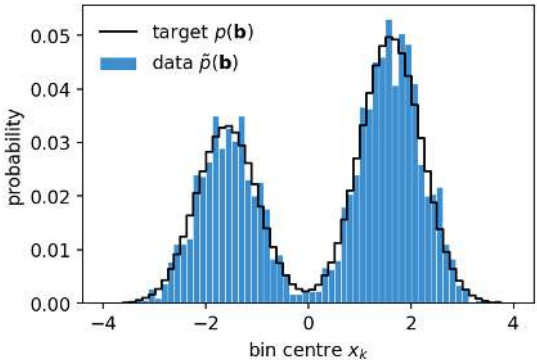



$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \sum_{\mathbf{a}} |\langle \mathbf{b}, \mathbf{a} | \psi(\boldsymbol{\theta}) \rangle|^2 = \langle \mathbf{b} | \hat{\rho}_{\text{meas}} | \mathbf{b} \rangle$$

a mixture of Born distributions, one for each ancilla string **a**

- Every distribution is $|\psi_{\mathbf{b}}|^2$ for some pure state, but a circuit of fixed depth reaches only some of them.
- Ancilla qubits take part in the circuit but are not measured, so the Born probabilities are summed over their bits.
- The model becomes a mixture of Born distributions.

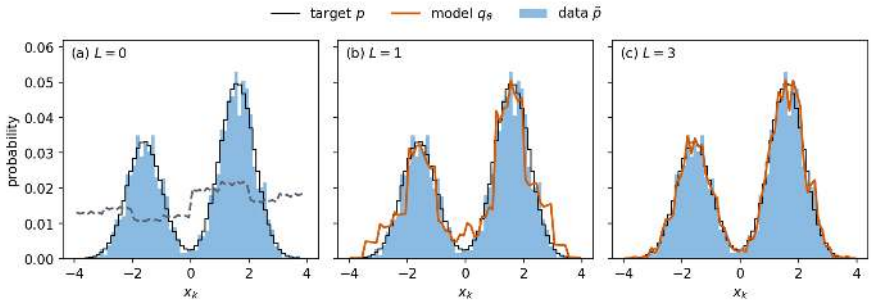
# Two Peaks Make the Bits Strongly Correlated




implementation: see notebooks Bars, empirical distribution of the data. Line, exact bin probabilities of the target.

- Mixture of two Gaussians, weights 0.4 and 0.6, centres $\mp 1.6$, width 0.6 on $[-4, 4]$. On $N = 6$ qubits the 64 bins have $w = 0.125$.
- The learner sees $|\mathcal{D}| = 3000$ values. Their distance to the target, $d_{\mathrm{TV}}(p, \tilde{p}) = 0.065$, sets the statistical error of the data.
- Left peak near bin 19, 010011, right near bin 44, 101100. Their leading bits are opposite, so the bits are strongly correlated.

# Three Entangling Layers Fit Peaks and Tails

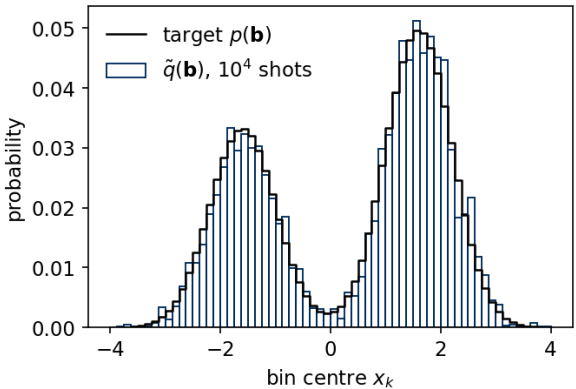



implementation: see notebooks

Trained model (line) and target for (a) the product circuit, (b) $L = 1$, (c) $L = 3$.

- $\hat{R}_z \hat{R}_y$ on every qubit alternate with $L$ $\widehat{\mathrm{CZ}}$ chains, Adam on the negative log-likelihood.
- The product model misses both peaks. One layer places them, three layers follow peaks and tails.

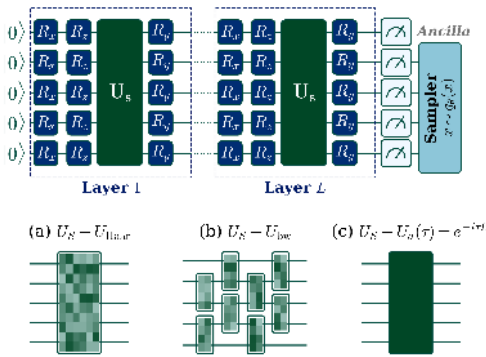
# Samples Are Judged by Total Variation Distance




implementation: see notebooks Bars, frequencies of 10000 measured bit strings of the $L = 3$ circuit. Line, exact target.

- On a device the circuit is used and tested through its samples.
- The histogram reproduces both peaks with their weights and widths.
- $d_{\mathrm{TV}}$ to the target is 0.051, close to 0.055 of the exact model probabilities.
- The KL divergence from samples is infinite once a populated bin gets no shot, so samples are judged by $d_{\mathrm{TV}}$.

# Only Rotations Train Around a Fixed Scrambler



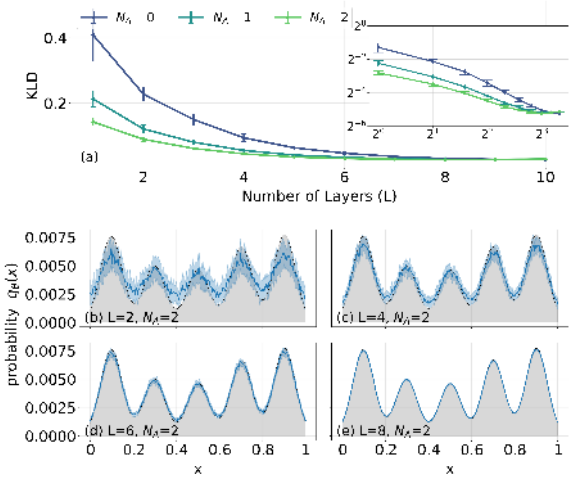

Trainable rotations on every qubit around a **fixed** entangling unitary $\hat{U}_{\mathrm{s}}$, the scrambler,

$$\hat{V}(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} \hat{R}_y^{(\ell)}\, \hat{U}_{\mathrm{s}}\, \hat{R}_z^{(\ell)} \hat{R}_x^{(\ell)}.$$

- Only the $3NL$ single-qubit angles are trained, the entangler is calibrated once.
- Unmeasured ancillas make the output a mixture of Born distributions.
- Three entanglers are compared to find which property sets the model quality, (a) Haar-random unitary, (b) brickwork of random two-qubit gates of depth $K$, (c) evolution $e^{-i\tau\hat{H}}$ of a qubit chain.

Figure: M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

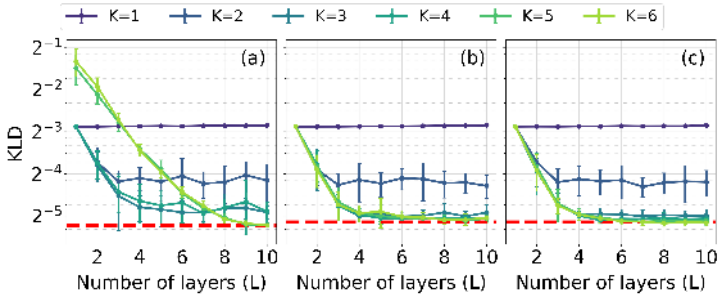
# More Layers Refine a Haar-Scrambled Model




- A uniformly drawn unitary maps $|0\rangle^{\otimes N}$ to a typical random state, nearly maximally entangled across every cut ("Random States in Three Facts").
- Target with five Gaussian peaks, $N = 8$.
- (a) KL divergence against the number of layers $L$. Traced-out ancillas ($N_A = 1, 2$) help most for shallow circuits.
- (b)–(e) Learned (blue) against target (grey), $L = 2, 4, 6, 8$.

Figure: M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

# Page-Level Entanglement Suffices for the Scrambler

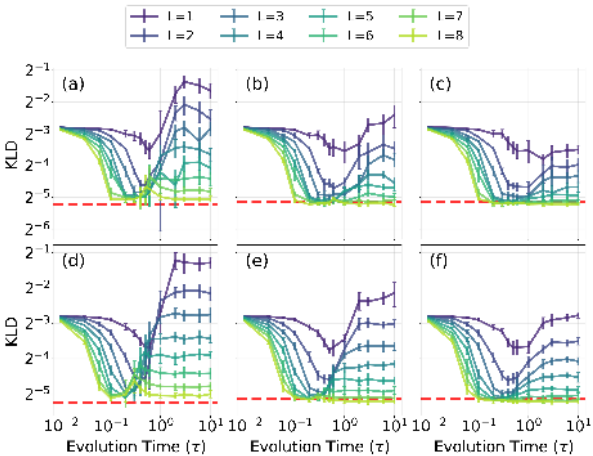



Brickwork of random two-qubit gates of depth $K$, KL divergence against $L$ for $N_A = 0, 1, 2$, Haar level red dashed.

- At $K = 1$ the model saturates far above the Haar level. At depth of order $N/2$ every curve collapses onto it.
- The model quality is set by how close the entangler comes to the Page value, not by the microscopic origin of the scrambler.

Figure: M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

# Analog Chain Evolution Works as a Scrambler



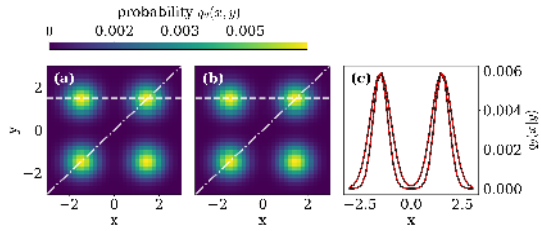

- Scrambler: evolution for a time $\tau$ under the Ising chain (top) or the XX chain in a field (bottom), native on an analog simulator.
- At short $\tau$ it barely entangles and training fails for every $L$.
- Around $\tau \sim 0.5$ (inverse coupling) $S_{\mathrm{vN}}$ approaches the Page value and the KL divergence drops to the Haar level for $L \geq 5$.
- Both chains are integrable, the trained site-dependent rotations break this.

Figure: M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

# A Trainable Hamiltonian Learns a 2D Distribution

Every layer evolves for a time $\tau$ under its own trainable Hamiltonian, $(3N+1)L$ parameters,

$$\hat{V}(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} e^{-i\tau \hat{H}^{(\ell)}}, \qquad \hat{H}^{(\ell)} = J^{(\ell)} \sum_i \hat{X}_i \hat{X}_{i+1} + \sum_i \left( h_{x,i}^{(\ell)} \hat{X}_i + h_{y,i}^{(\ell)} \hat{Y}_i + h_{z,i}^{(\ell)} \hat{Z}_i \right).$$




Four Gaussian peaks over two variables, $N = 10$ with two ancillas, $L = 10$, 310 parameters. (a) Target, (b) learned, (c) slices, target black, model red.

- All four peaks are reproduced with their weights. Learning a distribution becomes a variational Hamiltonian problem.

Figure: M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

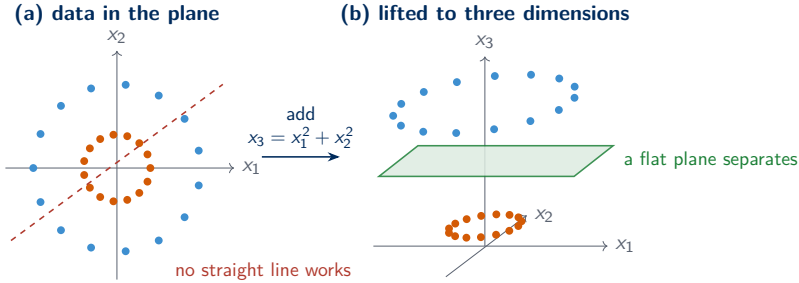
# 8

# Circuits as Classifiers: Quantum Kernels



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# A Feature Map Makes Rings Linearly Separable




- No straight line in the plane separates two rings.
- A new coordinate $x_3 = x_1^2 + x_2^2$ lifts the inner ring low and the outer ring high, and a flat plane separates them.
- After a suitable **feature map** $\mathbf{x} \mapsto \phi(\mathbf{x})$ into a higher dimension, a hyperplane separates the classes.

# An SVM Needs Only Inner Products

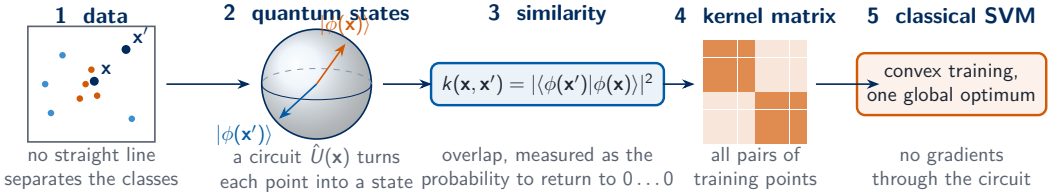
A support-vector machine (SVM) finds the hyperplane with the widest margin between classes $y \in \{-1, +1\}$, using only the **kernel**

$$k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}'), \qquad f(\mathbf{x}) = \operatorname{sign}\Big( \sum_{i=1}^{M} \underbrace{\alpha_i}_{\text{learned}} \; y_i \; \underbrace{k(\mathbf{x}_i, \mathbf{x})}_{\text{similarity to training point } i} + b \Big).$$

- Only the $M \times M$ kernel matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ of the training points is needed, never the feature vectors.
- Finding the weights $\alpha_i$ is a convex problem with one global optimum, solved by standard software.
- The kernel measures similarity, large for points the feature map sends to nearby vectors.

C. Cortes and V. Vapnik, Mach. Learn. **20**, 273 (1995), doi:10.1007/BF00994018

# Quantum States Serve as Feature Vectors



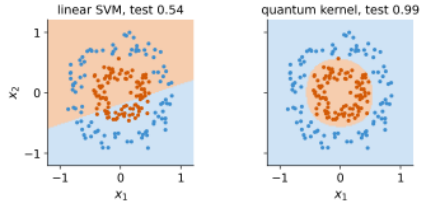

- A circuit $\hat{U}(\mathbf{x})$ maps each data point to a state $|\phi(\mathbf{x})\rangle$ of dimension $2^N$, never written down.
- The device measures the similarity of two states, a classical SVM does the rest.
- **Fixed kernel**: the feature map $\hat{U}(\mathbf{x})$ is chosen in advance, only the SVM is trained, a convex problem.
- **Trainable kernel**: the free angles of $\hat{U}(\mathbf{x};\boldsymbol{\theta})$ are tuned to make points of one class similar, then the SVM is solved.

V. Havlíček *et al.*, Nature **567**, 209 (2019), doi:10.1038/s41586-019-0980-2 M. Schuld and N. Killoran, PRL **122**, 040504 (2019), doi:10.1103/PhysRevLett.122.040504

# A Product Quantum Kernel Separates the Rings

Each coordinate rotates its own qubit, and the overlap of two such states factorises,

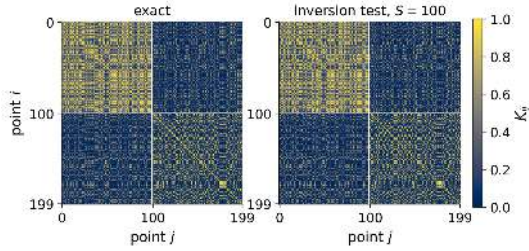
$$|\phi(\mathbf{x})\rangle = \hat{R}_y(\pi x_1)|0\rangle \otimes \hat{R}_y(\pi x_2)|0\rangle, \qquad k(\mathbf{x}, \mathbf{x}') = \cos^2\frac{\pi(x_1 - x_1')}{2}\,\cos^2\frac{\pi(x_2 - x_2')}{2}.$$




implementation: see notebooks

- Two noisy rings, 200 training and 200 test points. Test accuracy 0.54 for a linear SVM, 0.995 for the quantum-kernel SVM.
- This small kernel is easy to compute classically. It illustrates the mechanism without a quantum advantage.

# Kernel Entries Are Estimated from Shots

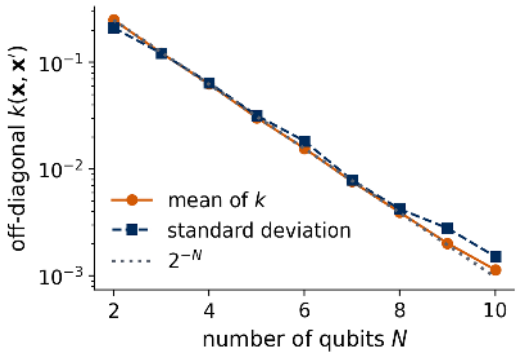



implementation: see notebooks

Training points sorted by class. Right, every entry estimated by the inversion test from $S = 100$ shots.

- Points of the same ring are similar (bright blocks), points of different rings are not.
- Each entry is the frequency of $0 \ldots 0$ after $\hat{U}(\mathbf{x}_j)^\dagger \hat{U}(\mathbf{x}_i)$, with statistical error $\sqrt{k(1-k)/S}$. The test accuracy stays at 0.995.
- Each pair needs its own circuit, $M(M-1)/2 \approx 2 \times 10^4$ for 200 training points, so the cost grows quadratically with the data set.

# Deep Feature Maps Make Kernels Concentrate




implementation: see notebooks Kernel values for 400 random pairs of points with a deep data encoding ($L = N$ layers of $\hat{R}_z\hat{R}_y$ and a $\widehat{\mathrm{CZ}}$ chain).

- A feature map that scrambles the data over the whole register sends different points to nearly orthogonal states.
- Off-diagonal kernel values shrink as $2^{-N}$, the overlap of two random states, and the kernel matrix approaches the identity.
- Resolving the overlaps needs about $10^5$ shots per entry at $N = 10$. The same concentration of random states causes barren plateaus ("Random States in Three Facts").
- A useful feature map keeps similar points similar.

S. Thanasilp *et al.*, Nat. Commun. **15**, 5200 (2024), doi:10.1038/s41467-024-49287-w

# 9

# Circuits as Classifiers: Data Re-uploading

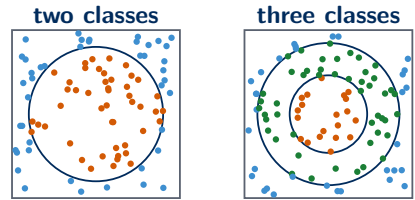


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# A Single Qubit as a Classifier

## Task

Points $\mathbf{x} = (x_1, x_2) \in [-1, 1]^2$ with class labels. Two classes: inside or outside a circle of radius $\sqrt{2/\pi}$, equal areas, guessing gives 50%. Three classes: disc, ring, outside. Learn a circuit that returns the class of a new point.

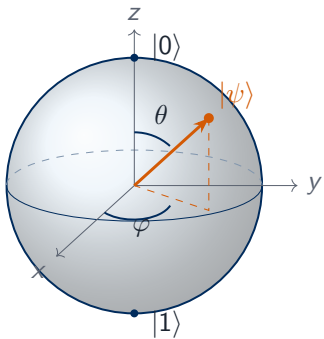



inside: class 1, outside: class 0 disc, ring, outside

- The smallest quantum model: *one* qubit. How much can it learn?
- Three ingredients: what a qubit state is, how a point enters the qubit, how a class is read out.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

# One Qubit: State and Data Encoding



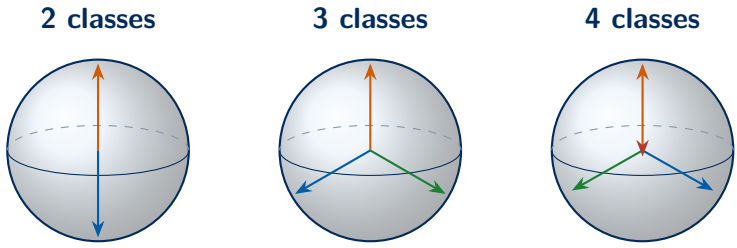

- **State.** $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$, a point on the Bloch sphere. Rotations $\hat{R}_y$, $\hat{R}_z$ move it, $P(0) = \cos^2\frac{\theta}{2}$.
- **Data encoding.** A point $\mathbf{x} = (x_1, x_2, 0)$ sets rotation angles with trainable $\boldsymbol{\theta}, \mathbf{w}$,

$$|\psi(\mathbf{x})\rangle = \hat{U}(\boldsymbol{\theta} + \mathbf{w} \circ \mathbf{x})|0\rangle,$$
$$\hat{U}(\boldsymbol{\phi}) = \hat{R}_z(\phi_3)\hat{R}_y(\phi_2)\hat{R}_z(\phi_1).$$

- The classifier is a *map from the plane to the sphere*.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

# Classes Are Label States on the Bloch Sphere




- Each class $c$ is a *label state* $|\psi_c\rangle$, the label states as far apart as possible.
- Two classes at the poles, three classes 120° apart on a great circle, four on a tetrahedron.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

# Prediction and Loss for Any Number of Classes

- Fidelity of the encoded point with each label state, measured as a frequency on a device,

$$F_c(\mathbf{x}) = |\langle \psi_c | \psi(\mathbf{x}) \rangle|^2, \qquad \text{prediction} \quad \hat{y}(\mathbf{x}) = \arg\max_c F_c(\mathbf{x}).$$

- Loss: average infidelity of each training point with its own label state,

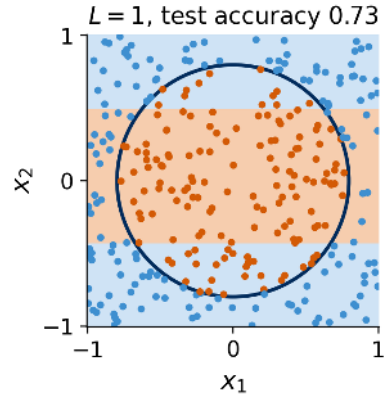
$$\mathcal{L}(\boldsymbol{\theta}, \mathbf{w}) = \frac{1}{M} \sum_{m=1}^{M} \left[ 1 - F_{y_m}(\mathbf{x}_m) \right].$$

- Two classes with label states $|0\rangle, |1\rangle$: $F_1 = P(1 \,|\, \mathbf{x})$, predict class 1 if $P(1 \,|\, \mathbf{x}) > \frac{1}{2}$. The decision boundary is mapped onto the equator.

- Three classes: label states $120^\circ$ apart, pairwise $F = 1/4$. Training pulls each region of the plane to its own label state.

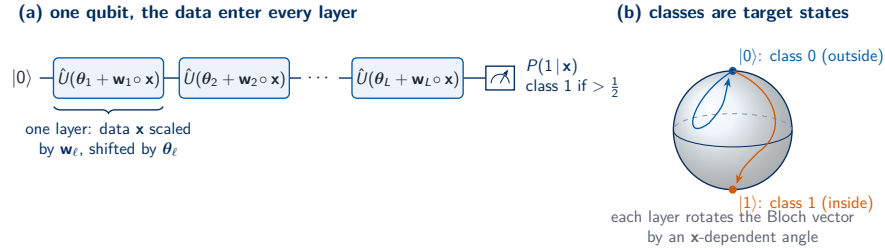
# One Rotation of One Qubit Draws a Band

- With one rotation, $P(1\,|\,\mathbf{x})$ depends on $x_2$ only through $\sin^2\left((\theta_2 + w_2 x_2)/2\right)$, so the boundary is a set of straight bands.
- The best band, $|x_2| < 0.58$, reaches a test accuracy of 0.76 on the circle task. One qubit seen once is too simple.




Two classes (orange inside, blue outside) and the circle. Shading, answer of a one-layer qubit classifier, which can only draw a band.

# Repeated Data Loading Builds Nonlinearity




$$|\psi(\mathbf{x})\rangle = \hat{U}(\boldsymbol{\theta}_L + \mathbf{w}_L \circ \mathbf{x}) \cdots \hat{U}(\boldsymbol{\theta}_1 + \mathbf{w}_1 \circ \mathbf{x})|0\rangle.$$

- Every layer loads the data again with its own weights $\mathbf{w}_\ell$ and offsets $\boldsymbol{\theta}_\ell$ ($w_{\ell,3}$ multiplies the padded zero).
- A layer acts like a neuron, an affine map of $\mathbf{x}$ followed by a nonlinear one, the rotation.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

# What a Classifier Can Draw Depends on Its Functions

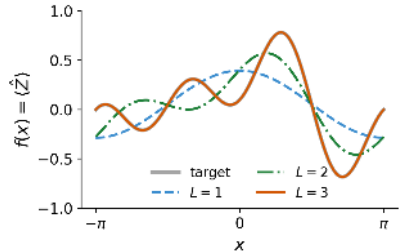
- The class of $\mathbf{x}$ is decided by $f(\mathbf{x}) = P(1 \mid \mathbf{x}) > \frac{1}{2}$. The decision boundary is the curve $f(\mathbf{x}) = \frac{1}{2}$.
- A model can only draw boundaries contained in its family of functions $f$. This family is its **expressive power**.
- Neural networks: with enough neurons any function, hence any boundary, can be represented (universal approximation). Which functions can one qubit produce?
- Fourier analysis answers this exactly. The *frequencies* of $f$ say how fast it can change with $\mathbf{x}$: low frequencies give smooth functions and simple boundaries, high frequencies give rapid changes and curved boundaries.

# Each Upload Adds One Fourier Frequency

- Take one input $x$, loaded by $e^{-ix\hat{Z}/2} = \mathrm{diag}\left(e^{-ix/2}, e^{ix/2}\right)$ between trainable gates $\hat{W}_\ell$.
- **One upload**: the amplitudes carry $e^{\mp ix/2}$. The output $f(x) = \langle\psi(x)|\hat{M}|\psi(x)\rangle$ multiplies an amplitude by a conjugate one, so $f = A + B\cos(x - \delta)$, a single frequency.
- ***L* uploads**: the phases add up, amplitudes contain $e^{-ikx/2}$ with $k = -L, \ldots, L$, and

$$f(x) = \sum_{\omega=-L}^{L} c_\omega \, e^{i\omega x}, \qquad c_{-\omega} = c_\omega^*,$$

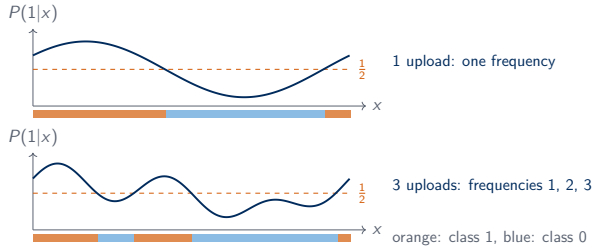
  a Fourier series up to frequency $L$, coefficients set by the $\hat{W}_\ell$.




implementation: see notebooks Target with frequencies 1, 2, 3. Only $L = 3$ uploads fit it exactly.

M. Schuld, R. Sweke and J. J. Meyer, PRA **103**, 032430 (2021), doi:10.1103/PhysRevA.103.032430

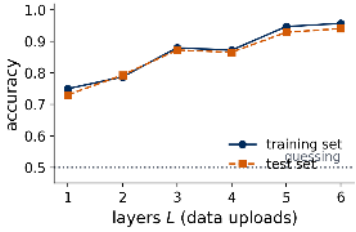
# A Fourier Series Draws the Decision Boundary



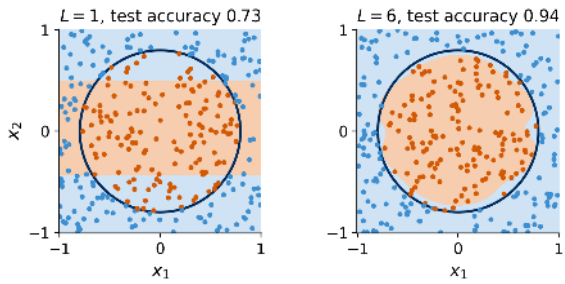

- The classifier compares $f(x) = P(1\,|\,x)$ with $\frac{1}{2}$. The boundary is where the curve crosses $\frac{1}{2}$.
- More uploads give more frequencies, more crossings, more complex class regions. For one input a single qubit can express any Fourier series.
- For several inputs a single qubit has too few parameters for all Fourier terms. Trainable data weights **w**, as in our circuit, restore universality.

Z. Yu *et al.*, NeurIPS 2022, arXiv:2205.07848 A. Pérez-Salinas *et al.*, PRA **104**, 012405 (2021), doi:10.1103/PhysRevA.104.012405

# A Single Qubit Is a Universal Classifier




implementation: see notebooks Circle task, 400 training and 4000 test points, best of four starts per $L$.

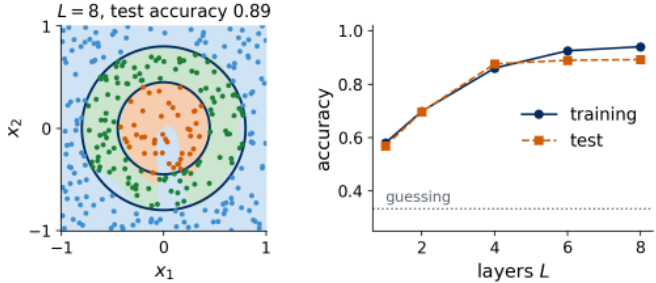



Learned classes (shading) against the true circle for $L = 1$ and $L = 6$.

- More uploads add higher frequencies and more complicated boundaries, and the test accuracy rises from 0.73 ($L = 1$, a trained band, slightly below the best band's 0.76) to 0.94 ($L = 6$).
- With trainable data weights and enough layers, one qubit approximates any bounded function of the data (Pérez-Salinas *et al.*), like a network with one hidden layer.

A. Pérez-Salinas *et al.*, PRA **104**, 012405 (2021), doi:10.1103/PhysRevA.104.012405

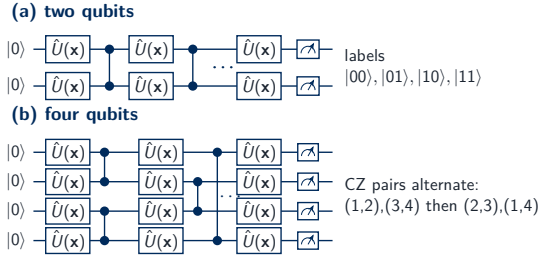
# One Qubit Separates Three Classes




implementation: see notebooks

- Disc, ring and outside, label states 120° apart (pairwise $F = 1/4$), prediction by the largest fidelity, loss = mean infidelity with the true label state.
- Test accuracy rises from 0.57 at $L = 1$ to 0.89 at $L = 8$, against 0.49 for always answering the largest class.

# More Qubits: More Label States, More Uploads




- Every qubit loads the data in every layer. A CZ after each layer (except the last) entangles the qubits.
- **Labels**: $N$ qubits have $2^N$ orthogonal basis states, so up to $2^N$ classes get perfectly distinguishable labels, $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ for four classes, read off in one shot.
- The paper also reads out single qubits and compares their reduced states with single-qubit label states.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

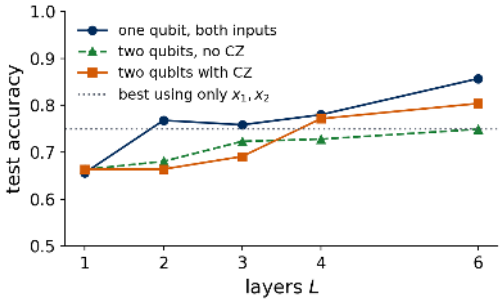
# Entanglement Helps When Data Must Be Combined

| problem (classes, input dim.) | 1 qubit | 2 qubits | | 4 qubits | |
|---|---|---|---|---|---|
| | | no ent. | ent. | no ent. | ent. |
| circle (2, 2D) | 0.94 | 0.96 | 0.96 | 0.96 | 0.96 |
| 3 circles (4, 2D) | 0.76 | 0.90 | 0.83 | 0.90 | 0.89 |
| annulus (3, 2D) | 0.76 | 0.79 | 0.82 | 0.87 | 0.96 |
| hypersphere (2, 4D) | 0.87 | 0.92 | 0.91 | 0.90 | **0.98** |

- Test accuracy with $L = 2$ layers (weighted-fidelity cost). More qubits help in every problem.
- At $L = 2$ entanglement helps the four-qubit annulus and sphere. With more layers most differences shrink, but for the 4D sphere entangled circuits stay ahead (two qubits at $L = 4$: 0.97 vs 0.93), as its class depends on all four coordinates together.

Values from Tables 1–4 of A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

# Entanglement Carries Information to the Read-out Qubit

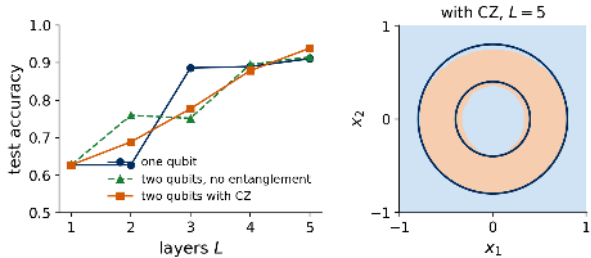



implementation: see notebooks 4D sphere $|\mathbf{x}|^2 < 4/\pi$ (class 1: 48% of the points), 1000 training and 4000 test points, best of four starts.

- Qubit 0 loads $(x_1, x_2)$, qubit 1 loads $(x_3, x_4)$, only qubit 0 is measured.
- **No CZ**: qubit 0 never learns $x_3, x_4$. It stays at 0.75, the best any rule using only $x_1, x_2$ can do.
- **With CZ**: information from qubit 1 reaches the read-out, and the accuracy passes that limit (0.80 at $L = 6$).
- One qubit loading all four inputs reaches 0.86. Entanglement is needed when the data are *spread over qubits*.

# Uploads Matter More Than Entanglement Here

- Annulus $0.4 < r < 0.8$ is class 1 (always answering 0 gives 0.63). Each layer uploads $\mathbf{x}$ into both qubits, $\hat{U}(\boldsymbol{\theta}_{\ell,q} + \mathbf{w}_{\ell,q} \circ \mathbf{x})$, $q = 0, 1$, then $\widehat{\mathrm{CZ}}$. Label states $|00\rangle$, $|11\rangle$.




implementation: see notebooks  best of six starts

- All models need several uploads to draw two boundaries. At $L = 5$: 0.94 with $\widehat{\mathrm{CZ}}$, 0.91 without, so here uploads matter more than entanglement.

A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

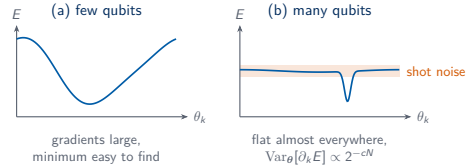
# 10

# Fundamental Limitations of Variational Quantum Circuits



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IN KRAKOW

# Deep Random Circuits Have Flat Landscapes




- Training often starts from angles drawn uniformly from $[-\pi, \pi]$.
- Along one angle every cost is $a + b\cos(\theta_k - c)$, so its derivative averages to zero, and the typical gradient size is set by its variance,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \left\langle (\partial_k C)^2 \right\rangle_{\boldsymbol{\theta}}.$$

- For deep random circuits this variance decays exponentially with $N$, a **barren plateau**, flat almost everywhere.
- A device must then resolve exponentially small gradients above the shot noise.

J. R. McClean *et al.*, Nat. Commun. **9**, 4812 (2018), doi:10.1038/s41467-018-07090-4

# Plateaus Make Shot Counts Grow Exponentially

A parameter-shift component from $M$ shots per setting has variance $\sigma^2/(2M)$, $\sigma^2$ the single-shot energy variance. It resolves the typical gradient when this equals $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]$,

$$M^\star = \frac{\sigma^2}{2\,\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]} \xrightarrow{\text{deep random circuit}} d - 1 = 2^N - 1.$$

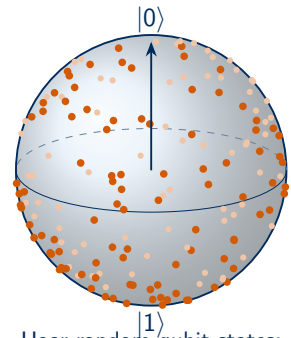
| | HEA, $N = 10$ | random, $N = 10$ | random, $N = 20$ | random, $N = 50$ |
|---|---|---|---|---|
| $M^\star$ | 58 | 1023 | $1 \times 10^6$ | $1 \times 10^{15}$ |

- A deep random circuit doubles the shot count for one gradient component with every qubit.
- An HVA angle enters $N - 1$ gates, and each of its $2(N - 1)$ shifted circuits needs only about 6.7 shots at $N = 10$.
- A better optimiser cannot remove a plateau, since every optimiser on a device must resolve energy differences above the shot noise.

implementation: see notebooks

# The Haar Measure Defines Uniformly Random States

- The **unitary group** $U(d)$: all $d \times d$ matrices with $\hat{U}^\dagger \hat{U} = \hat{I}$, every circuit on $N$ qubits, $d = 2^N$.
- The **Haar measure**, the uniform distribution on $U(d)$, is unchanged by multiplying with any fixed $\hat{V}$,
$$d\mu(\hat{U}) = d\mu(\hat{V}\hat{U}) = d\mu(\hat{U}\hat{V}).$$
- A **Haar-random state** $|\psi\rangle = \hat{U}|0\ldots0\rangle$ is the uniformly random unit vector of "Random States in Three Facts". Sampling: $d$ complex Gaussian amplitudes, normalised.
- One qubit: Haar states fill the Bloch sphere uniformly (right).

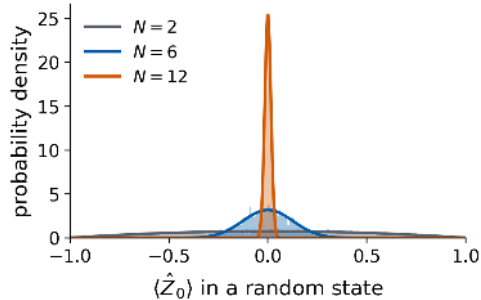


Haar-random qubit states: uniform on the Bloch sphere

F. Mezzadri, arXiv:math-ph/0609050 A. A. Mele, Quantum **8**, 1340 (2024), doi:10.22331/q-2024-05-08-1340

# On a High-Dimensional Sphere Almost Everything Is Near the Equator

- Pick a uniformly random point on the unit sphere in $\mathbb{R}^n$. One coordinate has density $\propto (1 - x_1^2)^{(n-3)/2}$ and variance $1/n$.
- For large $n$ that coordinate is almost surely close to 0: nearly all the sphere lies near the equator. This holds for every axis at once.
- Quantum states: $\langle \hat{Z}_0 \rangle$ of a random $N$-qubit state has variance $1/(2^N + 1)$. Each added qubit halves the variance, so the histogram narrows by a factor $\sqrt{2}$.




implementation: see notebooks $\langle \hat{Z}_0 \rangle$ in 4000 Haar-random states, $N = 2, 6, 12$.

# Lévy's Lemma: Smooth Quantities Are Almost Constant

Let $f(\psi)$ be any smooth quantity of a state, for example $f(\psi) = \langle\psi|\hat{O}|\psi\rangle$.

- $\eta$, how fast $f$ can change: $|f(\psi) - f(\phi)| \leq \eta \, \|\psi - \phi\|$. For an expectation value $\eta \leq 2\|\hat{O}\|$.
- $\epsilon$, the size of a deviation from the average that we call atypical.

$$\Pr\Big(\big|f(\psi) - \langle f\rangle_{\text{Haar}}\big| > \epsilon\Big) \;\leq\; 2\,e^{-C\,d\,\epsilon^2/\eta^2}, \qquad d = 2^N,\ C > 0 \text{ a constant.}$$

- The probability of any visible deviation falls *exponentially in the dimension* $d = 2^N$.
- Meaning: almost every random state gives practically the same value of every smooth quantity, its Haar average. Typical deviations are of order $2^{-N/2}$.
- Gradients of a cost are such quantities, which is the root of barren plateaus.

M. Ledoux, *The Concentration of Measure Phenomenon* (AMS, 2001), doi:10.1090/surv/089 P. Hayden, D. W. Leung and A. Winter, Commun. Math. Phys. **265**, 95 (2006), doi:10.1007/s00220-006-1535-6

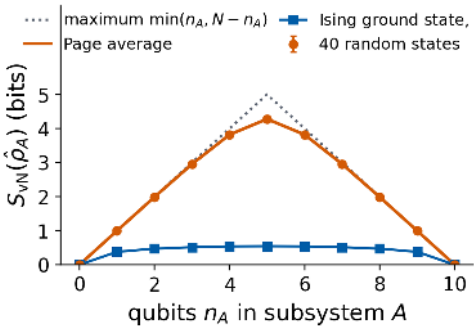
# Typical States Look Featureless Locally

By Lévy's lemma almost every random state is **typical**, its measurable properties equal to Haar averages up to errors exponentially small in $N$, as one microstate represents the microcanonical ensemble.

- **Locally featureless.** A subsystem $A$ much smaller than the rest has $\hat{\rho}_A = \mathrm{Tr}_B |\psi\rangle\langle\psi| \approx \hat{I}_A / d_A$. Every local observable gives $\mathrm{Tr}\, \hat{O}_A / d_A$, every Pauli string $\langle \hat{P} \rangle \approx 0$ (canonical typicality).
- **Information is global.** The state is pure, yet all its information sits in correlations among many qubits, invisible to local measurements.
- **Nearly orthogonal.** Two random states overlap by $|\langle\psi|\phi\rangle|^2 \approx 1/d = 2^{-N}$.
- **Consequence.** Deep random circuits produce states that all look alike locally, so changing one angle hardly changes a local cost.

S. Popescu, A. J. Short and A. Winter, Nat. Phys. **2**, 754 (2006), doi:10.1038/nphys444 S. Goldstein *et al.*, PRL **96**, 050403 (2006), doi:10.1103/PhysRevLett.96.050403

# Typical States Carry Near-Maximal Entanglement




implementation: see notebooks Entropy of $n_A$ qubits of an $N = 10$ chain, 40 Haar-random states, Page average and the critical Ising ground state.

D. N. Page, PRL **71**, 1291 (1993), doi:10.1103/PhysRevLett.71.1291

Average entanglement entropy of $n_A \leq N/2$ qubits in a random state (Page),

$$\langle S_{\mathrm{vN}}(\hat{\rho}_A)\rangle \approx n_A - \frac{d_A}{2 d_B \ln 2} \text{ bits},$$

with $d_A = 2^{n_A}$, $d_B = 2^{N-n_A}$.

- Nearly maximal, growing with the volume of $A$, with tiny fluctuations.
- Ground states of local Hamiltonians obey an area law, so they are atypical states in a tiny corner of Hilbert space.
- A circuit that reaches every state mostly produces typical states, far from such targets.

# Concentration Makes Gradient Variance Exponentially Small

Deep random circuits reproduce the first and second moments of the Haar measure (a *unitary 2-design*), so every expectation value concentrates,

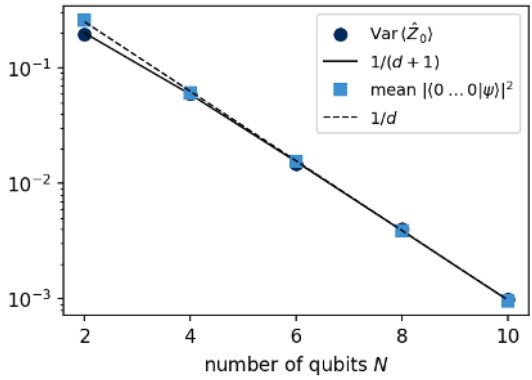
$$\mathrm{Var}\,\langle \hat{P}\rangle = \frac{1}{d+1} \approx 2^{-N} \quad (\hat{P} \neq \hat{I}), \qquad \big\langle |\langle 0\ldots 0|\psi\rangle|^2 \big\rangle = \frac{1}{d},$$

- $\langle \hat{Z}_0 \rangle$ is a difference of two sums of $d/2$ random weights.
- If all gates after the differentiated rotation act as a Haar-random unitary, a cost $C = \langle \hat{O} \rangle$ has

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \qquad \hat{O}_0 = \hat{O} - \frac{\mathrm{Tr}\,\hat{O}}{d}\,\hat{I}.$$

- The global cost $\hat{O} = \hat{I} - |0\ldots 0\rangle\langle 0\ldots 0|$ gives $\mathrm{Var} \approx 2^{-2N-1}$.
- The Ising energy, a sum of Pauli strings, gives $\mathrm{Var} \approx [(N-1)J^2 + Nh^2]/2^{N+1}$.

# Random-State Averages Halve with Every Qubit

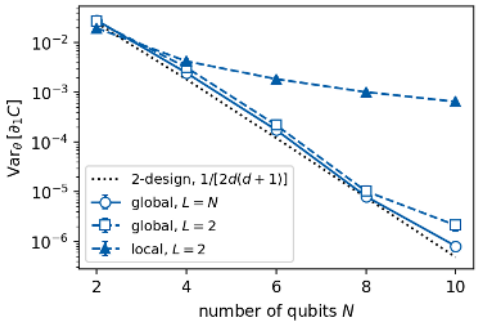



implementation: see notebooks

Variance of $\langle \hat{Z}_0 \rangle$ and mean of $|\langle 0 \ldots 0|\psi\rangle|^2$ over 2000 random states against $N$, lines $1/(d+1)$ and $1/d$, $d = 2^N$.

- Both halve with every added qubit.

# Local Costs Keep Larger Gradients




implementation: see notebooks Variance of $\partial C/\partial\theta_1$ over 300 random angle vectors for the HEA, against $N$, with the 2-design value of the global cost.

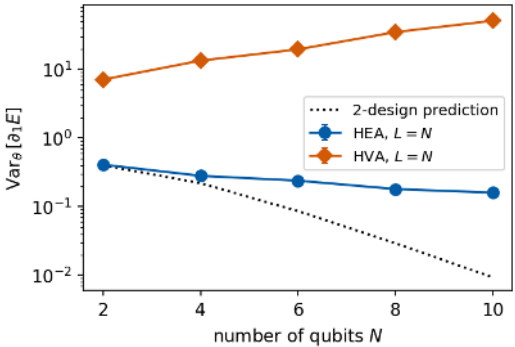
Both costs vanish only in $|0\ldots0\rangle$, the autoencoder costs with all $N$ qubits as trash,

$$C_{\mathrm{glob}} = 1 - |\langle 0\ldots0|\psi\rangle|^2, \qquad C_{\mathrm{loc}} = \frac{1}{N}\sum_q \frac{1-\langle \hat{Z}_q\rangle}{2}.$$

- The global cost decays as $2^{-cN}$, $c = 1.96$ at $L = N$ (2-design value) and $c = 1.81$ already at $L = 2$.
- The local cost at $L = 2$ sees only the light cone of the differentiated gate, and $N^2\,\mathrm{Var}$ stays constant.
- At $N = 10$ its variance is 311 times larger.

M. Cerezo *et al.*, Nat. Commun. **12**, 1791 (2021), doi:10.1038/s41467-021-21728-w

# The HVA Gradient Variance Grows with Size




implementation: see notebooks Variance of the Ising energy gradient at $h = J$ for the HEA and the HVA at depth $L = N$, with the 2-design value.

- The energy is a sum of terms on one or two neighbouring qubits.
- For the HEA at depth $N$ the gradient decays far more slowly than the 2-design curve at these sizes, since the circuit has only just connected the two chain ends.
- The HVA variance grows with $N$. One shared angle changes all $2N - 1$ terms together.
- HVA gates generate unitaries whose dimension, that of the dynamical Lie algebra, grows only polynomially with $N$.

M. Ragone *et al.*, Nat. Commun. **15**, 7172 (2024), doi:10.1038/s41467-024-49909-3 M. Larocca *et al.*, Nat. Rev. Phys. **7**, 174 (2025), doi:10.1038/s42254-025-00813-9

# Summary

1. **Learning models.** A circuit with classically tuned angles is a trainable model.
2. **Gradients.** Two shifted circuits give an exact derivative (SPSA: backup).
3. **VQE.** A small energy error does not certify the state, symmetry and entanglement do.
4. **Autoencoders.** A clean trash register means lossless compression.
5. **QCNN and QGRNN.** Pooling reads a string order, Trotter layers learn a Hamiltonian.
6. **Circuit autoencoder.** The Choi spectrum bounds the compression of a channel.
7. **Born machines.** A scrambler near the Page value suffices.
8. **Classifiers.** Kernels compare quantum states, re-uploading builds Fourier series.
9. **Limitations.** Global costs of deep random circuits have barren plateaus.

# Labs in Course Order

- **069** parametrised circuits, energy from bit strings, gradients.
- **069a** SPSA and Adam with few circuit runs.
- **070** VQE and the choice of ansatz, Ising chain.
- **070a** VQE phase diagram of the XXZ chain in a field.
- **070b** quantum autoencoder and anomaly detection.
- **070c** QCNN for the cluster-Ising chain.
- **070d** QGRNN, Hamiltonian learning.
- **070e** VQE for the XYZ chain, three ansätze.
- **070f** quantum circuit autoencoder.
- **071a** random states, Lévy's lemma, typicality.
- **071** barren plateaus, global and local costs.
- **072** Born machines for a continuous distribution.
- **073a** data re-uploading classifier.
- **073b** quantum kernels.
- **073** QUBO and QAOA, at home (lecture notes only).

# Reviews and Further Reading


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# 12

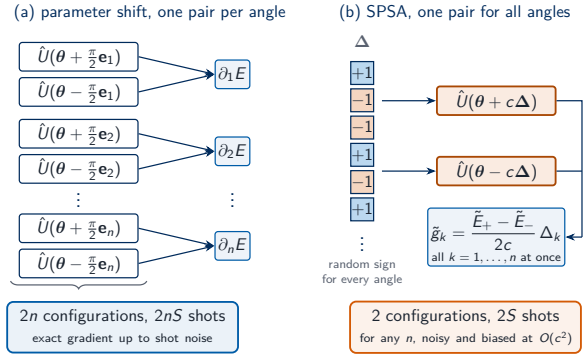
## Backup: Gradients from Few Circuit Runs (SPSA)



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# Circuit Configurations per Gradient

**Circuit configuration**: one set of angles, measured with $S$ shots in total (all settings).




Parameter shift moves one $\theta_k$ to $\theta_k \pm \frac{\pi}{2}$, others fixed. SPSA moves all angles at once, $\boldsymbol{\theta} \pm c\boldsymbol{\Delta}$, for any $n$.

J. C. Spall, IEEE TAC **37**, 332 (1992), doi:10.1109/9.119632 K. Mitarai *et al.*, PRA **98**, 032309 (2018), doi:10.1103/PhysRevA.98.032309

# Two Energies Estimate the Whole Gradient (SPSA)

**Update $t = 0, 1, 2, \ldots$**

1. Draw $\Delta_k = \pm 1$, probability $\frac{1}{2}$, independently for every angle.
2. Measure $\tilde{E}_\pm = \tilde{E}(\boldsymbol{\theta} \pm c_t \boldsymbol{\Delta})$ from two configurations.
3. Estimate all components $(1/\Delta_k = \Delta_k)$,
$$\tilde{g}_k = \frac{\tilde{E}_+ - \tilde{E}_-}{2c_t}\,\Delta_k .$$
4. Step $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - a_t\,\tilde{\mathbf{g}}$ with gains
$$a_t = \frac{a}{(t+1+A)^{0.602}},\quad c_t = \frac{c}{(t+1)^{0.101}}.$$

$c_t$: perturbation size, $a_t$: step size, $a, c, A$: chosen constants. Exponents from Spall (1998).

**Average over $\boldsymbol{\Delta}$ is the gradient**

$$\frac{E_+ - E_-}{2c} = \underbrace{\textstyle\sum_j \Delta_j\,\partial_j E}_{\text{slope along } \boldsymbol{\Delta}} + O(c^2)$$

$$\big\langle \tilde{g}_k \big\rangle_{\boldsymbol{\Delta}} = \textstyle\sum_j \underbrace{\langle \Delta_k \Delta_j \rangle}_{\delta_{kj}}\,\partial_j E + O(c^2)$$

$$= \partial_k E + O(c^2)$$

- Bias $O(c^2)$ from third derivatives. Used in VQE on superconducting hardware.
- Variance $\sum_{j\neq k} (\partial_j E)^2 + \sigma_{\tilde{E}}^2/(2c^2)$ grows with $n$.

J. C. Spall, IEEE TAC **37**, 332 (1992), doi:10.1109/9.119632 J. C. Spall, IEEE TAES **34**, 817 (1998), doi:10.1109/7.705889 A. Kandala *et al.*, Nature **549**, 242 (2017), doi:10.1038/nature23879

# Adam Driven by SPSA Gradients

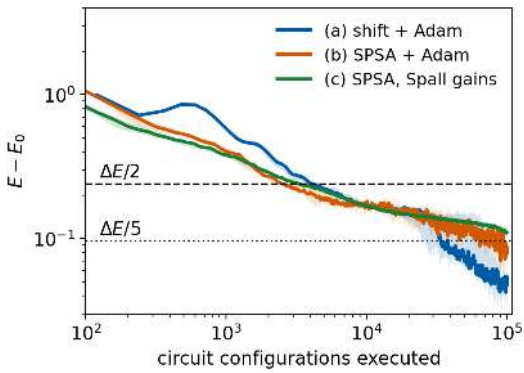
- Adam takes $\tilde{\mathbf{g}}$ in place of the exact gradient, at 2 configurations per update.
- Each angle keeps running averages of $\tilde{g}_k$ and $\tilde{g}_k^2$, step $\eta \times (\text{first})/\sqrt{\text{second}}$.
- Random sign flips of $\tilde{g}_k$ cancel in the first average, so the step follows the mean gradient with size close to $\eta$.
- Fixed $\eta$ and $c$ replace the gains $a_t, c_t$.

## Task

Critical Ising chain, $N = 6$, hardware-efficient circuit with $n = 60$ angles started near $|0 \ldots 0\rangle$, $S = 2000$ shots per configuration. Count configurations until $E - E_0 \leq \Delta E/2$ and $\Delta E/5$, with the gap $\Delta E = E_1 - E_0$. Since $E - E_0 \geq \Delta E(1 - F)$, these guarantee fidelities $F = |\langle E_0|\psi\rangle|^2 \geq \frac{1}{2}$ and $\geq 0.8$. Optimisers: (a) parameter shift + Adam, (b) SPSA + Adam, (c) SPSA with Spall's gains.

D. P. Kingma and J. Ba, arXiv:1412.6980 J. C. Spall, IEEE TAC **37**, 332 (1992), doi:10.1109/9.119632

# SPSA Cheaper Far from the Minimum, Shift Near It




implementation: see notebooks $E - E_0$ exact (simulation), never seen by the optimisers. Lines show the median of 5 seeds, bands the smallest to largest value.

Median configurations (updates) until

| | $\Delta E/2$ | $\Delta E/5$ |
|---|---|---|
| (a) | 4 080 (34) | 33 840 (282) |
| (b) | 2 520 (1 260) | 45 920 (22 960) |
| (c) | 3 280 (1 640) | not within $10^5$ |

- SPSA needs far more updates, each costing 2 configurations against $2n = 120$; shots $= S \times$configurations.
- Near the minimum the gradient is small, the SPSA shot noise $\propto 1/(c^2 S)$ dominates and the parameter shift wins.